

Joint Latency Control for Synchronized Multi-Receiver Audio: Model, Loss, and the Structure of the Optimal Controller

August 20, 2026

Abstract

This document states and solves a control problem for synchronized audio streaming to N receivers over a local network. The problem has one loss with four terms: mean latency (linear), cross-receiver synchronization error (quadratic), buffer underrun (an indicator), and warp-rate roughness (quadratic). The disturbance is one stochastic process per receiver: the per-sample delivery delay, modeled as baseline wander plus stationary diffusion plus a pause process. The document derives the structure of the optimal controller. The derivation is explicit about every simplification. Each simplification gets a statement, a reason, a quantified defect, and a telemetry monitor. The result: the optimal controller factors into per-receiver linear loops (a specific second-order law with damping ratio $1/\sqrt{2}$) and a slow scalar program that positions the common latency level. The latency target is not an input of the problem. It appears in the solution as a dynamic state variable of the controller.

Contents

1	Introduction	3
1.1	The system in one page	3
1.2	The three goals are one problem	3
1.3	How to read this document	3
2	The System Model	4
2.1	Timelines, samples, and schedules	4
2.2	The buffer and underrun	5
2.3	Measurement: two feedback architectures	5
3	The Delay Process	6
3.1	The three components	6
3.2	What the model does not assume	7
3.3	Levels, not increments	7
4	The Loss	8
4.1	Policies	8
4.2	The loss functional	8
4.3	Modeling choices, stated openly	9
5	What Each Loss Term Can See	9
5.1	Linear and quadratic terms need only moments	9
5.2	The indicator term is not determined by moments	10
5.3	Matching the marginal is still not enough for event structure	10
5.4	Where the temporal law does and does not matter	11

6	Exact Reductions	12
6.1	From time averages to stationary expectations	12
6.2	Mean and difference coordinates	12
6.3	Superposition	13
7	The Quadratic Theory	13
7.1	The closed algebra	13
7.2	Spectral evaluation	14
7.3	The fast loop is a linear-quadratic regulator	14
7.4	Speaker-space diagonalization	16
7.5	Closed-loop transfer functions and the component ledger	17
7.6	Integral action	17
8	The Indicator Term	18
8.1	Exact rewriting	18
8.2	Evaluation under the component model	19
8.3	The simplification chain	20
8.4	Shape of the risk curve	22
9	Composition: the Two Solvers and the Free Direction	23
9.1	The free direction	23
9.2	The decomposition theorem	24
9.3	The level program	25
9.4	Per-receiver tilts	25
9.5	Moving the level: glides and asymmetry	26
9.6	What remains of the general problem	28
10	The Resulting Controller	28
10.1	Layer 1: estimators (the belief state)	28
10.2	Layer 2: the level program	29
10.3	Layer 3: fast loops (unchanged)	29
10.4	Layer 4: event hygiene	29
10.5	Self-monitoring	29
11	Register of Simplifications	29
12	Limits of the Result	29
A	Imported Results	31
A.1	Campbell–Mecke formula	31
A.2	Rice’s formula	31
A.3	Gaussian tail independence	32
A.4	Other imports	32
B	Notation	32

1 Introduction

1.1 The system in one page

One sender streams audio over a local network to N receivers. Each receiver drives one loud-speaker. The sender packetizes samples and sends them with their stream timestamps. Each receiver holds arrived samples in a buffer and plays them out through a resampler. The resampler can play slightly faster or slower than nominal rate. This rate adjustment is called *warp*. Warp is the only actuator in the system.

Three things can go wrong.

- (a) **Latency.** Audio plays late relative to capture. For interactive sources (video, instruments) this is a direct cost.
- (b) **Desynchronization.** Two receivers play the same sample at different times. The listener hears smeared or displaced sound.
- (c) **Underrun.** A receiver needs a sample that has not arrived. It must insert silence. This is an audible dropout.

A fourth cost is internal: warp changes pitch. Fast warp changes are audible. Slow warp offsets are not.

These four costs pull in different directions. Lower latency means a smaller buffer, which means more underruns. Tighter synchronization means stiffer control, which means rougher warp. The purpose of this document is to treat the four costs as one optimization problem and to derive the controller that this problem implies.

1.2 The three goals are one problem

Historically the three goals were handled by three mechanisms: a fixed latency target chosen by hand (latency), independent per-receiver controllers holding that target (synchronization), and a safety margin inside the target (underrun). This document shows that the three mechanisms are the correct *shape* of the solution, but that the target must not be an input. The optimization produces the target as an output. The target becomes an internal, slowly moving state of the controller, computed from measured statistics of the network.

1.3 How to read this document

Sections 2–4 state the problem: the physical model, the stochastic process, and the loss. These sections contain no approximations beyond three explicitly flagged identifications: an $O(10^{-4})$ actuator rescaling (theorem 2.1), the margin-queue identification to one packet period (Remark 2.2), and the zero-drain ramp idealization (theorem 3.1). Section 5 proves which statistics of the process each loss term can see. This section explains *why* the solution splits into two different kinds of mathematics. Section 6 performs exact coordinate changes. Section 7 solves the quadratic part of the problem with linear-quadratic machinery. Section 8 handles the underrun term. Every simplification lives in a numbered environment (**S1**, **S2**, ...) with a quantified defect. Section 9 composes the two solutions and proves the decomposition result. Section 10 states the resulting controller as an algorithm. Section 11 collects all simplifications in one table. Section 12 lists what the derivation does *not* cover.

Two style rules apply throughout. First: every symbol is defined before use, and section B collects the recurring ones. Second: every claim is either proven here, or imported with an explicit citation and marked as imported. There are no silent steps.

2 The System Model

2.1 Timelines, samples, and schedules

Convention 2.1 (Units). All times are in seconds. Stream position is measured in seconds of audio, not in sample counts. This removes the sample rate F_s from all formulas. To convert to sample units, multiply seconds of audio by F_s .

Definition 2.1 (Stream and emission). The audio stream is indexed by *stream time* $s \in \mathbb{R}$: the position inside the audio signal, in seconds of audio. The sender emits (captures and transmits) stream position s at wall time $\tau(s)$. For an ideal sender clock, $\tau(s) = s + \text{const}$; we absorb the constant and write $\tau(s) = s + c_A(s)$, where c_A is the sender clock error, a slowly varying function with $|\dot{c}_A| \leq 10^{-4}$ (clock rates differ from nominal by at most tens of ppm).

Definition 2.2 (Arrival). Stream position s becomes available in receiver i 's buffer at wall time

$$a_i(s) = \tau(s) + D_i(s), \quad (1)$$

where $D_i(s) \geq 0$ is the *delivery delay* of receiver i for stream position s . The delay D_i covers everything between emission and availability: packetization, network transit, queueing, and any receiver-side processing before the sample is playable. The delay processes $(D_1, \dots, D_N)$ are the stochastic disturbance of the problem. Section 3 models them.

Definition 2.3 (Playout schedule and warp). Receiver i chooses a strictly increasing *playout schedule* $p_i(s)$: stream position s plays at wall time $p_i(s)$. Let $s_i(t)$ denote the inverse: the stream position scheduled at wall time t , so $p_i(s_i(t)) = t$. The *warp* is the fractional playout rate offset

$$u_i(t) := \frac{ds_i}{dt}(t) (1 + \dot{c}_{B,i}) - 1, \quad (2)$$

where $c_{B,i}$ is receiver i 's clock error function, defined like c_A in Definition 2.1, with the same ppm-scale rate bound. In words: $u_i > 0$ means receiver i consumes the stream faster than nominal; $u_i < 0$ means slower. The warp is bounded by the resampler: $|u_i| \leq u_{\max}$, with $u_{\max} = 5 \times 10^{-3}$ in the implementation.

Definition 2.4 (End-to-end latency). The *latency* of receiver i at wall time t is the age of the sample it is scheduled to play:

$$L_i(t) := t - \tau(s_i(t)). \quad (3)$$

Proposition 2.1 (Latency dynamics). *The latency obeys*

$$\dot{L}_i(t) = w_i(t) - u_i(t), \quad |w_i(t)| \leq 3 \times 10^{-4}, \quad (4)$$

where $w_i := (\dot{c}_{B,i} - \dot{c}_A)/(1 + \dot{c}_{B,i})$ is the relative clock drift between receiver i and the sender.

Proof. Differentiate eq. (3): $\dot{L}_i = 1 - \tau'(s_i(t)) \dot{s}_i(t) = 1 - (1 + \dot{c}_A) \dot{s}_i$. By eq. (2), $\dot{s}_i = (1 + u_i)/(1 + \dot{c}_{B,i})$. Therefore

$$\dot{L}_i = 1 - \frac{(1 + \dot{c}_A)(1 + u_i)}{1 + \dot{c}_{B,i}} = \frac{\dot{c}_{B,i} - \dot{c}_A - u_i(1 + \dot{c}_A)}{1 + \dot{c}_{B,i}}.$$

Define $w_i := (\dot{c}_{B,i} - \dot{c}_A)/(1 + \dot{c}_{B,i})$ and absorb the factor $(1 + \dot{c}_A)/(1 + \dot{c}_{B,i}) = 1 + O(10^{-4})$ into the definition of u_i (a relative rescaling of the actuator by at most 10^{-4} , far below actuator resolution). Both clock rates are within 10^{-4} of nominal, so $|w_i| \leq 2 \times 10^{-4}/(1 - 10^{-4}) < 3 \times 10^{-4}$. $\square$

Remark 2.1 (What eq. (4) says). Latency is an integrator driven by clock drift and steered by warp. The controller cannot set latency directly. It can only slide latency around at up to $u_{\max} = 5 \text{ ms per second}$. This bound is the single most important quantitative fact in this document. It reappears in Simplification S3, where it decouples the underrun term from the fast control loop, and in section 9.5, where it limits how fast the latency level can be raised.

2.2 The buffer and underrun

Definition 2.5 (Underrun). Receiver i is *in underrun* at wall time t when the sample it is scheduled to play has not arrived:

$$\text{underrun}_i(t) \iff a_i(s_i(t)) > t. \quad (5)$$

Convention 2.2 (Silence convention). During underrun the schedule keeps advancing and the receiver outputs silence for the missing samples. The schedule is not paused and not reset. Missing audio is counted by the loss (section 4); it is not compensated later.

Proposition 2.2 (Underrun is a delay-versus-latency comparison). *For all i, t :*

$$\text{underrun}_i(t) \iff D_i(s_i(t)) > L_i(t). \quad (6)$$

Proof. By eq. (1), $a_i(s_i(t)) > t$ iff $\tau(s_i(t)) + D_i(s_i(t)) > t$ iff $D_i(s_i(t)) > t - \tau(s_i(t)) = L_i(t)$ by eq. (3). $\square$

Definition 2.6 (Margin). The *margin* of receiver i at wall time t is

$$m_i(t) := L_i(t) - D_i(s_i(t)). \quad (7)$$

Underrun at t is exactly $m_i(t) < 0$.

Remark 2.2 (Margin and buffer depth). When arrivals are in stream order, the buffer of receiver i at time t holds exactly the samples s with $s_i(t) \leq s$ and $a_i(s) \leq t$, and the buffered duration equals the margin up to the local variation of D_i over one buffer length. Packet reordering breaks exact equality but the identification “margin $\approx$ buffered audio duration” is accurate to one packet period. We use the word *margin* for the model quantity and *queue depth* for the implementation measurement.

2.3 Measurement: two feedback architectures

The controller observes the system through one of two measured outputs. Both exist in the implementation; the choice is per stream.

Definition 2.7 (Latency measurement, “e2e” backend). The receiver measures

$$y_i(t) = L_i(t) + n_i(t), \quad (8)$$

where n_i is measurement error. Its dominant component is the error of the clock synchronization between sender and receiver: computing $t - \tau(s)$ requires comparing the two clocks. On hosts disciplined by the same time service, n_i is microseconds. On an undisciplined host, n_i can drift by milliseconds over minutes; this drift enters the loop as a false disturbance. Section 7.5 quantifies the consequence.

Definition 2.8 (Margin measurement, “niq” backend). The receiver measures its own buffered duration:

$$y_i(t) = m_i(t). \quad (9)$$

This measurement needs no cross-host clock comparison. It is exact up to packet quantization. Its cost: the measured signal now contains the delay process, so network disturbances drive the control loop (section 7.5).

3 The Delay Process

This section models the disturbance $(D_1, \dots, D_N)$. The model is the result of measurement, not of convenience; each component corresponds to a distinct physical mechanism, and the telemetry estimates each component separately.

3.1 The three components

Assumption 3.1 (Additive decomposition). For each receiver i ,

$$D_i(s) = B_i(t_s) + \varepsilon_i(s) + P_i(s), \quad t_s = \tau(s), \quad (10)$$

where B_i is the *baseline*, ε_i is the *diffusion*, and P_i is the *pause component*, defined in Assumptions 3.2 to 3.4 below.

Assumption 3.2 (Baseline). $B_i(t)$ is differentiable and slow: its derivative is bounded, $|\dot{B}_i| \leq \bar{\beta}$ with $\bar{\beta}$ of order 10^{-5} (tens of microseconds per second), and its power spectrum is concentrated below the control bandwidth κ defined in section 7.3. Physically: propagation and processing floor, thermal drift of clocks and interface timing, slow route or load changes.

Assumption 3.3 (Diffusion). $\varepsilon_i(s)$ is stationary, mean-zero, and mixing (correlations decay; there is a horizon beyond which values are almost independent). It has variance σ_i^2 and power spectral density $S_{\varepsilon,i}(\omega)$. Its tail is light: there are constants x_0, C such that $\mathbb{P}(\varepsilon_i > x) \leq Ce^{-x/x_0}$ for all $x > 0$. Physically: interrupt coalescing, scheduler granularity, cross-traffic queueing at sub-millisecond scale. Measured values in the reference deployment: σ_i from tens of microseconds up to about one millisecond on wired hosts. The upper end matters: at an aggressive latency level of order 10ms the margin is only a few multiples of σ_i , so the tail of ε_i is a first-class contribution to underrun risk, not a negligible one (section 8.2).

Assumption 3.4 (Pause component). Delivery to receiver i stops entirely during *pause episodes*. Episode k starts at wall time $T_{i,k}$ and lasts $H_{i,k}$ seconds (the *height*); delivery resumes at $T_{i,k} + H_{i,k}$. Stream content emitted during the episode is delivered late (after resume), recovered by forward error correction, or lost; the split between these outcomes is a property of the episode mechanism. The pairs $(T_{i,k}, H_{i,k})$ form a stationary marked point process with intensity λ_i (episodes per second) and mark law F_i (the Palm distribution of heights; $\bar{F}_i := 1 - F_i$). Clustering of episode times is allowed and is a measured property, not a parameter. Physically: radio retraining, roaming, bulk interference, CPU stalls. Measured heights range from milliseconds (wired hosts) to 10^2 – 10^3 milliseconds (radio hosts).

Proposition 3.1 (Pause ramp shape). *Under Assumption 3.4 and the zero-drain idealization below, during episode k the pause component is a descending ramp: for stream positions s with $\tau(s) \in [T_{i,k}, T_{i,k} + H_{i,k}]$,*

$$P_i(s) = T_{i,k} + H_{i,k} - \tau(s), \quad (11)$$

and $P_i(s) = 0$ outside episodes. In particular $\max_s P_i(s) = H_{i,k}$ during episode k : the height equals both the duration of the episode and the peak extra delay.

Zero-drain idealization: *withheld content is treated as arriving at the instant delivery resumes. In reality the burst drains over the time of Assumption 3.5(a) (milliseconds on a local network); the ramp is exact up to that drain time. This idealization is carried as part of Simplification S2's defect accounting (section 8.3).*

Proof. A sample emitted at $\tau(s)$ inside the episode cannot arrive before delivery resumes at $T_{i,k} + H_{i,k}$. If it is delivered at all, it arrives in the burst of withheld content just after resume — at the resume instant, under the idealization — so its extra delay is $T_{i,k} + H_{i,k} - \tau(s)$, which

falls linearly from $H_{i,k}$ (for the sample emitted at episode start) to 0 (at episode end). Samples emitted outside episodes have no withheld path, so $P_i = 0$. $\square$

Definition 3.1 (Recovery time). The *recovery time* $\rho_{i,k}$ of episode k is the time from delivery resume until the margin of receiver i is back at its pre-episode value.

Assumption 3.5 (Recovery mechanisms). No instant recovery is assumed. Two mechanisms bound $\rho_{i,k}$:

- (a) **Redelivery burst (fast)**. When the withheld content was queued upstream, it arrives in a burst at the link rate once delivery resumes. The burst refills the buffer directly: $\rho_{i,k} \approx H_{i,k} \cdot R_{\text{stream}}/R_{\text{link}}$ (stream bit rate over link capacity), milliseconds on a local network. This is the measured behavior of radio dropouts in the reference deployment: the instantaneous queue depth spikes by approximately the episode height right after resume.
- (b) **Warp-limited restoration (slow)**. When content was lost, or when the playout schedule slipped during the episode (a stalled playout, or a controller that converted the transient into latency), the deficit can only be worked off by warp: $\rho_{i,k} \leq \chi_{i,k}/u_{\max}$, where $\chi_{i,k} = \min(H_{i,k}, m_i)$ is the margin the episode consumed. With $u_{\max} = 5 \times 10^{-3}$, a consumed margin of 10 ms takes up to 2 s to restore; 100 ms takes up to 20 s.

Which mechanism applies is a property of the episode and of the implementation path; ρ is treated as a measured per-episode mark. The interval after an episode during which the margin is still depressed is a window of raised vulnerability; section 8.2 prices a following episode inside this window using the measured gap law.

Assumption 3.6 (Cross-receiver structure). The joint process $(D_1, \dots, D_N)$ is stationary. Components decompose along the network path tree: each of B, ε, P splits into a part common to all receivers (sender host, shared switch), parts shared by subtrees, and idiosyncratic parts. No independence across receivers is assumed anywhere. Measured fact: pause episodes co-occur across receivers at rates two orders of magnitude above the independence prediction.

3.2 What the model does not assume

The model does not assume Poisson pause times (clustering is measured). It does not assume any specific height law F_i (heavy or multimodal laws are allowed; heights are bounded in practice by session supervision, so all moments of D_i exist). It does not assume Gaussian diffusion (only light tails and mixing). It does not assume independence across receivers. It does not assume instant recovery after an episode (Assumption 3.5), and it does not assume that the diffusion tail is negligible at the operating margin.

3.3 Levels, not increments

Classical jitter metrics are built from *increments*: differences of consecutive delays. The following elementary fact shows that increments cannot drive a buffer model, and it calibrates what the white-noise terms in section 7 may and may not represent.

Proposition 3.2 (Increment sums telescope). *Let $\eta(n)$, $n \in \mathbb{Z}$, be any stationary sequence (per-packet delay levels) with variance $\sigma_\eta^2 < \infty$, and let $\xi(n) = \eta(n) - \eta(n-1)$. Then for every M ,*

$$\sum_{n=1}^M \xi(n) = \eta(M) - \eta(0), \quad \text{Var}\left[\sum_{n=1}^M \xi(n)\right] \leq 4\sigma_\eta^2,$$

so the long-run variance rate of the partial sums is zero: $\lim_{M \rightarrow \infty} \text{Var}[\sum_1^M \xi]/M = 0$.

Proof. The sum telescopes term by term. The variance of a difference of two variables, each with variance σ_η^2 , is at most $4\sigma_\eta^2$ by the Cauchy–Schwarz inequality. Dividing a bounded quantity by M gives zero in the limit. $\square$

Corollary 3.3. *A stationary delay process contributes no Brownian (accumulating) noise to the buffer. The buffer responds to the delay level directly: a delay change of x seconds moves the margin by $-x$ seconds, once, and the margin recovers when the delay recovers. Any model that drives the buffer with white noise of nonzero intensity implicitly assumes a nonstationary (wandering) delay level. In this document, white noise terms therefore represent only the baseline wander B_i and measurement noise; the stationary diffusion ε_i enters through its spectrum and the pause component P_i through the underrun term.*

4 The Loss

4.1 Policies

Definition 4.1 (Causal policy). A *policy* π is a rule that chooses every receiver’s warp $u_i(t)$ as a function of observations available at time t : the receiver’s own measured output y_i up to t (Definitions 2.7 and 2.8), its own arrival history, and any messages received from the sender by time t . Messages from receivers reach the sender after a one-way network delay, and sender broadcasts reach receivers after another; any closed information loop through the sender takes at least a round trip. The information structure is therefore distributed: no receiver sees another receiver’s state directly, and the sender sees receiver reports with delay.

4.2 The loss functional

Let ω denote one realization of the joint disturbance: the delay processes of all receivers and the clock drifts. All randomness lives in ω .

Definition 4.2 (The loss). For a policy π , define

$$J(\pi) = \mathbb{E}_\omega \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Lambda(t, \omega) \, dt \right], \quad (12)$$

with the instantaneous loss rate, evaluated on the realization,

$$\Lambda(t, \omega) = \alpha \bar{L}(t) + \frac{q}{N} \sum_{i=1}^N (L_i(t) - \bar{L}(t))^2 + \frac{c}{N} \sum_{i=1}^N \mathbf{1}[m_i(t) < 0] + \frac{r}{N} \sum_{i=1}^N \dot{u}_i(t)^2, \quad (13)$$

where $\bar{L}(t) = \frac{1}{N} \sum_i L_i(t)$ is the instantaneous fleet mean latency and $\alpha, q, c, r > 0$ are weights.

The four terms, in words:

- (i) **Latency (linear).** Every second of average latency costs α per second of operation. Latency is bad in proportion to its size; there is no threshold and no target.
- (ii) **Synchronization (quadratic).** The instantaneous variance of latency across receivers, on this realization. This is the physical desynchronization the listener hears. It contains no reference value: it compares receivers with each other only.
- (iii) **Underrun (indicator).** The fraction of receivers currently playing silence for lack of data. By theorem 2.2 this is a property of delay versus latency; by Convention 2.2 silence duration is what accumulates.

- (iv) **Warp roughness (quadratic).** Fast warp changes are audible pitch transients. A constant warp is free: only the derivative is charged. This choice has a structural consequence used in section 9: the loss puts zero price on any constant warp offset, hence zero price on holding any fixed latency level.

Remark 4.1 (Units). Let u denote one loss unit. Then Λ has units u/s and

$$[\alpha] = u s^{-2}, \quad [q] = u s^{-3}, \quad [c] = u s^{-1}, \quad [r] = u s.$$

Two combinations will matter: α/c has units s^{-1} (a rate — it will equal the deviation’s occupation density at the optimum, section 9.3; for the pause component this is an episode exceedance rate), and $(q/r)^{1/4}$ has units s^{-1} (a bandwidth — it will be the fast loop’s natural frequency in section 7.3). Dimensional consistency of both is a useful check on the derivation.

4.3 Modeling choices, stated openly

Three choices in Definition 4.2 are choices, not necessities.

- (a) **Mean over receivers in the latency term.** The maximum over receivers is an alternative. With the synchronization term active, receivers sit within a fraction of a millisecond of each other, so mean and maximum differ by less than the resolution of any latency preference. The mean gives cleaner algebra.
- (b) **Linear latency cost.** Any increasing function of latency is admissible. Linearity makes α interpretable (the price of one second of latency) and, in section 9.3, makes the operating point depend on the single ratio α/c .
- (c) **Underrun counted by time, symmetrically across receivers.** The indicator charges silence duration. It does not separately charge the *number* of dropouts, and it does not model listener adaptation. A count-based or risk-averse variant changes section 8; section 12 discusses where.

Remark 4.2 (What is absent: session restarts). The implementation supervises sessions and restarts them on gross failures (unbounded queue growth, tolerance violations). A restart is a loud, discrete event that this loss does not model beyond the silence it causes. The derivation therefore optimizes normal operation; supervision remains a separate safety layer. See section 12.

5 What Each Loss Term Can See

The four loss terms differ in which statistics of the disturbance they depend on. This section makes that precise. The result explains the architecture of the whole solution: three terms live inside a closed “moment algebra” that linear-quadratic theory solves, and one term — the indicator — provably lives outside it.

5.1 Linear and quadratic terms need only moments

Proposition 5.1 (First moments suffice for linear terms). *Let x be a random vector with mean μ , and let $g(x) = a^\top x$ be linear. Then $\mathbb{E}[g(x)] = a^\top \mu$ depends on the law of x only through its first moment.*

Proof. Linearity of expectation. □

Proposition 5.2 (Second moments suffice for quadratic terms). *Let x be a random vector with mean μ and covariance Σ , and let $g(x) = x^\top Q x$. Then*

$$\mathbb{E}[g(x)] = \text{tr}(Q\Sigma) + \mu^\top Q\mu, \tag{14}$$

which depends on the law of x only through its first two moments.

Proof. Write $x = \mu + z$ with $\mathbb{E}[z] = 0$, $\mathbb{E}[zz^\top] = \Sigma$. Then $\mathbb{E}[x^\top Qx] = \mu^\top Q\mu + 2\mu^\top Q\mathbb{E}[z] + \mathbb{E}[z^\top Qz]$, and $\mathbb{E}[z^\top Qz] = \mathbb{E}[\text{tr}(Qzz^\top)] = \text{tr}(Q\Sigma)$ by linearity of the trace and of expectation. $\square$

For stationary processes passed through linear time-invariant systems, “second moments” means the power spectral density; section 7.2 gives the exact statement. The consequence: for the latency, synchronization, and warp terms, two disturbance processes with equal first and second moments produce *identical* loss under every linear policy. Everything else about the disturbance is invisible to these terms.

5.2 The indicator term is not determined by moments

Proposition 5.3 (Moment insufficiency, marginal form). *There exist two stationary processes with equal mean and equal autocovariance function whose stationary underrun fractions at the same margin differ by more than eighty orders of magnitude.*

Proof. Take the pause process of Assumption 3.4 with constant intensity $\lambda = 10^{-2}\text{s}^{-1}$, deterministic height $H \equiv 0.5\text{s}$, non-overlapping episodes, and no other components. By theorem 3.1, during an episode the level descends linearly from H to 0 at unit rate, so the fraction of time the level exceeds $m = 0.4\text{s}$ is, by theorem 5.5 below, $\lambda \mathbb{E}(H - m)^+ = 10^{-2} \times 0.1 = 10^{-3}$. Its stationary mean is $\mu_P = \lambda \int_0^H (H - x) dx = \lambda H^2/2 = 1.25 \times 10^{-3}\text{s}$, and its stationary second moment is $\lambda \int_0^H (H - x)^2 dx = \lambda H^3/3 = 4.17 \times 10^{-4}\text{s}^2$, giving variance $\approx 4.15 \times 10^{-4}\text{s}^2$, standard deviation $\sigma_P \approx 20.4\text{ms}$. Its autocovariance function is some fixed function $C(\cdot)$.

Now take the Gaussian process with the same mean μ_P and the same autocovariance C (one exists for any valid autocovariance). Its stationary exceedance of $m = 0.4\text{s}$ is $\bar{\Phi}((0.4 - 0.00125)/0.0204) = \bar{\Phi}(19.5) < 10^{-84}$, using the Mills-ratio bound $\bar{\Phi}(x) \leq \varphi(x)/x = e^{-x^2/2}/(x\sqrt{2\pi})$. The two processes agree on all first and second moments and disagree on the underrun fraction by a factor exceeding 10^{80} . $\square$

Remark 5.1. The example is extreme by design, but the mechanism is the measured one: pause heights are hundreds of standard deviations of the bulk. Any moment-matched Gaussian surrogate concludes that underruns never happen.

5.3 Matching the marginal is still not enough for event structure

A natural repair: transform a Gaussian process through a monotone map so that the *marginal* distribution matches exactly (a “trans-Gaussian” or Gaussian-copula model), and match the covariance as well. This fixes the exceedance *fraction* by construction. It still fails on the *time structure* of exceedances, and the failure is qualitative.

Proposition 5.4 (Sojourn shrinkage for Gaussian-based models). *Let X be a stationary, almost-surely differentiable Gaussian process with variance σ_0^2 and derivative variance σ_1^2 . The mean duration of one visit of X above level z satisfies*

$$\mu_{\text{soj}}(z) = \frac{\mathbb{P}(X > z)}{N^+(z)} = \frac{\sqrt{2\pi}\sigma_0^2}{\sigma_1 z} (1 + o(1)) \xrightarrow{z \rightarrow \infty} 0, \quad (15)$$

where $N^+(z)$ is the rate of upcrossings of level z . A monotone marginal transform changes the level scale but not the vanishing: high-level visits of any trans-Gaussian process built on a smooth (almost surely differentiable) Gaussian base become arbitrarily short. By contrast, the pause process of Assumption 3.4 has mean visit duration $\mathbb{E}[H - m \mid H > m]$ above margin m : a quantity of the order of the height spread, bounded away from zero.

Proof. Two imports, both collected in section A.2: Rice’s formula for the upcrossing rate, and the ergodic occupation–crossing ratio identity $\mu_{\text{soj}}(z) = \mathbb{P}(X > z)/N^+(z)$ (mean visit duration equals stationary exceedance probability divided by visit rate). Rice’s formula gives $N^+(z) = \frac{\sigma_1}{2\pi\sigma_0} e^{-z^2/2\sigma_0^2}$. The Mills ratio bound $\bar{\Phi}(x) = \frac{\varphi(x)}{x}(1 + O(x^{-2}))$ with $x = z/\sigma_0$ gives $\mu_{\text{soj}} = \bar{\Phi}(x) \frac{2\pi\sigma_0}{\sigma_1} e^{x^2/2} = \frac{1}{\sqrt{2\pi}x} \frac{2\pi\sigma_0}{\sigma_1} (1 + o(1)) = \frac{\sqrt{2\pi}\sigma_0^2}{\sigma_1 z} (1 + o(1))$. The pause statement is theorem 3.1: one episode with height $H > m$ stays above m for exactly $H - m$. Underlying both is the tail-independence property of the Gaussian copula (section A.3). $\square$

Remark 5.2 (Scope of the sojourn argument). Theorem 5.4 is an asymptotic statement ($z \rightarrow \infty$). At a *fixed* operating margin it supplies the mechanism and the direction, not a bound: how short the transformed process’s visits are at a given level depends on the transform. The operational claim — that marginal-matched Gaussian-based models mispredict event durations at deployed margins — rests on the exact fixed-level counterexample (theorem 5.3) for magnitudes and on the measured event-duration histograms of the telemetry, with the asymptotics pinning the reason.

5.4 Where the temporal law does and does not matter

The last two propositions justify modeling pauses explicitly. The following exactness result then draws a perhaps surprising boundary: for the *mean* underrun loss, only marginal pause statistics matter.

Proposition 5.5 (Campbell evaluation of the underrun fraction). *Under Assumption 3.4, suppose episodes of receiver i do not overlap, the margin is held at the constant value m throughout, and recovery is instant ($\rho = 0$; section 8 quantifies the error of all three suppositions). Then the long-run fraction of time in underrun caused by pauses alone is*

$$R_i(m) = \lambda_i \mathbb{E}(H_i - m)^+, \quad (16)$$

where the expectation is over the Palm mark law F_i . This holds for any stationary marked point process: clustering of episode times does not change the mean underrun fraction.

Proof. By theorem 3.1 and Definition 2.6, during episode k the margin is $m - P_i$, which is negative exactly while $P_i > m$, i.e., for duration $(H_{i,k} - m)^+$ (the ramp descends at unit rate). Episodes do not overlap, so total underrun time in $[0, T]$ is $\sum_{k: T_{i,k} \in [0, T]} (H_{i,k} - m)^+$ up to edge effects of two episodes. The Campbell–Mecke formula for stationary marked point processes (imported; statement and reference in section A.1) gives the expectation of this sum as $\lambda_i T \mathbb{E}(H_i - m)^+$. Dividing by T and letting $T \rightarrow \infty$ gives eq. (16); the limit holds almost surely under ergodicity by the same theorem’s ergodic form. $\square$

Remark 5.3 (The division of labor). Combining theorems 5.3 to 5.5:

- Under instant recovery, the *mean* pause-caused underrun loss at a fixed margin needs only the marginal exceedance structure (λ_i, F_i) . Clustering does not move it.
- With finite recovery times (Assumption 3.5) the mean loss gains a compound term: an episode arriving while the margin is still depressed does extra damage. This term depends on the *gap law* between consecutive episodes, so the temporal structure enters the mean loss through exactly one channel — gaps shorter than recovery times. Section 8.2 computes it.
- Beyond the mean, the temporal law also governs the variance of realized loss over finite horizons (hence how fast (λ_i, F_i) can be estimated and how wide the confidence bands of the level program are), the validity of Simplifications S3–S5, and any risk-averse variant of the loss.

This sharpens the reason for measuring gap structure: it prices the compound risk, it calibrates the trust in the estimated risk, and it validates the model.

6 Exact Reductions

This section performs coordinate changes that are exact, preceded by the one statistical convention everything later rests on (S1 — stated first so that its scope is visible, and itself carrying a defect like every other S-item). The output is a decomposition of the loss into parts that the quadratic machinery of section 7 can process, plus a precise identification of the one term that resists.

6.1 From time averages to stationary expectations

Simplification 1 (S1: stationary ergodic regime). **Statement.** Within one operating regime, the closed-loop system under a fixed policy is stationary and ergodic: the time average in eq. (12) equals the expectation of Λ under the stationary law,

$$J(\pi) = \mathbb{E}[\Lambda(0, \omega)] \quad (\text{stationary law of the closed loop}).$$

Role. Converts trajectory optimization into optimization of a stationary distribution. Every later formula computes stationary expectations.

Defect. Real networks switch regimes: a host moves from wired to radio, a radio environment degrades, a route changes. Statistics estimated in one regime are wrong in the next. The size of the defect is the loss incurred during adaptation: (regime switch rate) $\times$ (adaptation time) $\times$ (excess loss during adaptation). The adaptation time of the estimators is a design parameter (section 10); the derivation does not hide it.

Monitor. Change detection on the estimator inputs; the telemetry flags disagreement between short-window and long-window estimates of $(\sigma_i, \lambda_i, F_i)$.

6.2 Mean and difference coordinates

Definition 6.1 (Fleet coordinates). Define the fleet mean and the deviations

$$\bar{L}(t) = \frac{1}{N} \sum_{i=1}^N L_i(t), \quad \tilde{L}_i(t) = L_i(t) - \bar{L}(t), \quad \sum_i \tilde{L}_i(t) = 0,$$

and the same for warp: $\bar{u} = \frac{1}{N} \sum_i u_i$, $\tilde{u}_i = u_i - \bar{u}$.

Proposition 6.1 (The loss in fleet coordinates). *The pointwise loss eq. (13) equals*

$$\Lambda = \underbrace{\alpha \bar{L} + r \dot{\bar{u}}^2}_{\text{mean coordinate}} + \underbrace{\frac{q}{N} \sum_i \tilde{L}_i^2 + \frac{r}{N} \sum_i \dot{\tilde{u}}_i^2}_{\text{difference coordinates}} + \underbrace{\frac{c}{N} \sum_i \mathbf{1}[m_i < 0]}_{\text{couples both}}. \quad (17)$$

Proof. The latency and synchronization terms are already in coordinate form by Definition 6.1. For the warp term, expand $\dot{u}_i = \dot{\bar{u}} + \dot{\tilde{u}}_i$:

$$\frac{1}{N} \sum_i \dot{u}_i^2 = \frac{1}{N} \sum_i (\dot{\bar{u}} + \dot{\tilde{u}}_i)^2 = \dot{\bar{u}}^2 + \frac{2\dot{\bar{u}}}{N} \sum_i \dot{\tilde{u}}_i + \frac{1}{N} \sum_i \dot{\tilde{u}}_i^2 = \dot{\bar{u}}^2 + \frac{1}{N} \sum_i \dot{\tilde{u}}_i^2,$$

because $\sum_i \tilde{u}_i = 0$ identically, hence $\sum_i \dot{\tilde{u}}_i = 0$. The indicator term depends on $m_i = \bar{L} + \tilde{L}_i - D_i(s_i(t))$, which contains both coordinates; it does not decompose. $\square$

Proposition 6.2 (Dynamics decouple). *From theorem 2.1, by averaging and subtracting:*

$$\dot{\bar{L}} = \bar{w} - \bar{u}, \quad \dot{\tilde{L}}_i = \tilde{w}_i - \tilde{u}_i.$$

The mean coordinate is driven only by mean drift and mean warp; each difference coordinate only by its own drift and warp deviations.

Proof. Linearity of eq. (4) in (w_i, u_i) and of the averaging map. $\square$

Remark 6.1 (The obstacle, stated precisely). After theorems 6.1 and 6.2, the problem would split into $1 + (N-1)$ independent problems — one for the mean, one per difference direction — *except* for the indicator term, which evaluates the sum $\bar{L} + \tilde{L}_i$ against the delay realization. Everything in section 8 exists to handle this one term; the result (section 9) is that after quantified approximations it attaches only to the mean coordinate.

6.3 Superposition

Proposition 6.3 (Component superposition). *Under any linear time-invariant policy, every closed-loop signal (latency error, margin, warp) is a sum of its responses to B_i , ε_i , P_i , drift w_i , and measurement noise n_i separately.*

Proof. The closed loop of a linear plant and a linear policy is a linear map of its inputs; Assumption 3.1 writes the disturbance as a sum. $\square$

This licenses the accounting of section 7.5: each loss term is evaluated component by component. Cross terms between components appear only in second moments of sums; Simplification 9 (S9) treats them.

7 The Quadratic Theory

This section solves the part of the problem that lives in the closed algebra of linear dynamics, quadratic costs, and second moments. The solution machinery is classical; the derivation is nevertheless complete, because the specific constants matter later.

7.1 The closed algebra

Three facts, jointly, make the quadratic part solvable in closed form.

Proposition 7.1 (Closure). *(i) A linear map of a Gaussian vector is Gaussian (imported; standard property of Gaussian measures).*

(ii) The expectation of a quadratic form depends only on the first two moments (theorem 5.2).

(iii) The minimizer of a quadratic cost under linear dynamics is linear state feedback (proven for our system in theorem 7.3 below).

Consequently: starting from linear dynamics, quadratic costs, and a disturbance described by second moments, every object generated by the optimization (optimal law, closed loop, cost value) stays inside a family parameterized by matrices. No path-space computation is needed.

Remark 7.1 (Where Gaussianity actually matters). For the fully observed problem eq. (19) below, the completion-of-squares argument of theorem 7.3 never uses Gaussianity: the linear law is optimal among *all* causal state-feedback laws for any mean-zero white noise of finite intensity, because the quadratic value function ansatz closes regardless of the noise law. Gaussianity earns its keep one level up: when the controlled error must be inferred from a corrupted measurement,

the linear estimator is optimal among all causal estimators only for Gaussian corruption. The real assumption of the fast layer is therefore about *measurement*, and Simplification 10 states it as such.

(Simplifications are numbered by their position in the register of section 11, not by order of appearance: S10, S12, and S13 arise inside the machinery and composition chapters, S2–S9 in the indicator chapter.)

Simplification 10 (S10: full-information law on the raw measurement). **Statement.** The fast layer applies the full-information optimal law of theorem 7.3 with the measured output substituted for the true error — no filtering stage — and stays in the linear time-invariant class.

Role. Keeps the fast layer a two-gain law; avoids carrying an estimator design through the derivation. Under exact error observation the restriction is vacuous (Remark 7.1).

Defect. Two parts, by backend. (a) Under the latency measurement, the corruption n_i enters the loop and moves true latency with transfer $-\kappa^2/\mathcal{D}(s)$ (theorem 7.6); the exposure is $\int |\kappa^2/\mathcal{D}(j\omega)|^2 S_n(\omega) d\omega/2\pi$, which also upper-bounds what an optimal filtering stage could recover. On time-disciplined hosts n_i is microseconds and the forfeit is negligible; on an undisciplined host n_i drifts by milliseconds and no linear filter repairs it — the correct response is switching that stream’s backend, not a filter. (b) Under the margin measurement, the deviation itself drives the loop, and its distribution is far from Gaussian (the pause component): a nonlinear law can strictly beat the linear one. The event-gated machinery of the implementation (detection, freezing, supervision) is exactly such a nonlinear element, added outside this derivation; its benefit is bounded by the pause leakage terms of theorem 7.6.

Monitor. Innovation histogram and spectrum of the loop; measured S_n level per host.

7.2 Spectral evaluation

Proposition 7.2 (Output spectra of linear systems). *Let x be stationary with power spectral density $S_x(\omega)$, let h be the impulse response of a stable linear time-invariant system with transfer function $H(s)$, and let $y = h * x$. Then y is stationary with*

$$S_y(\omega) = |H(j\omega)|^2 S_x(\omega), \quad \text{Var}[y] = \int_{-\infty}^{\infty} |H(j\omega)|^2 S_x(\omega) \frac{d\omega}{2\pi}. \quad (18)$$

Proof. The autocovariance of y is $R_y(\tau) = \mathbb{E}[y(t)y(t+\tau)] = \iint h(a)h(b) R_x(\tau+a-b) da db$. Taking Fourier transforms in τ and using the convolution theorem twice gives $S_y = H(j\omega)\overline{H(j\omega)} S_x$. Integrating the spectral density gives the variance (Bochner / Wiener–Khinchin, imported). $\square$

Convention 7.1 (Transform notation). In spectral and transfer-function contexts, $H(s)$ denotes a generic transfer function and s the Laplace variable; capital letters ($E, U, W, Y, M, N_i, \tilde{D}$) denote transforms of the corresponding time-domain signals. Neither use collides with stream position s (Definition 2.1) or episode heights $H_{i,k}$ (Assumption 3.4), which never appear inside transforms.

7.3 The fast loop is a linear-quadratic regulator

Consider one difference coordinate (section 7.4 shows all $N-1$ of them are identical copies). Write e for the coordinate’s latency error, u for its warp, $v = \dot{u}$ for the control input, and let white noise of intensity σ_w^2 drive e (per theorem 3.3, this white noise represents baseline-wander and measurement-noise components; the stationary diffusion enters separately through theorem 7.2 and the transfer functions of section 7.5). The system and cost are

$$\frac{d}{dt} \begin{pmatrix} e \\ u \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}}_A \begin{pmatrix} e \\ u \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_B v + \begin{pmatrix} \sigma_w \dot{W} \\ 0 \end{pmatrix}, \quad \text{cost rate } qe^2 + rv^2. \quad (19)$$

Simplification 12 (S12: white-noise surrogate for the loop-driving disturbance). **Statement.** The disturbance driving the fast-loop design is modeled as white noise of intensity σ_w^2 , although the physical components it stands for — the wander of B_i (Assumption 3.2: bounded derivative, not integrated white noise) and measurement error — have non-flat spectra.

Role. Yields the closed-form Riccati and Lyapunov results below.

Defect. Two distinct effects. For the *gains*: the optimal linear law against a colored disturbance augments the design with a disturbance model; for the dominant mismatch here — the near-DC drift component — that augmentation is exactly the integral action of section 7.6, which the implementation carries, so the residual gain defect is confined to mid-band spectral shape. For the *evaluations*: the variance formulas of theorem 7.4 err by $\int |H(j\omega)|^2 (S_{\text{true}}(\omega) - \sigma_w^2) d\omega/2\pi$ per output, a measurable quantity.

Monitor. Flatness of the loop-innovation spectrum within the bandwidth κ .

Proposition 7.3 (Riccati solution). *The average-cost optimal law for eq. (19) is*

$$v = \kappa^2 e - \sqrt{2}\kappa u, \quad \kappa := \left(\frac{q}{r}\right)^{1/4}, \quad (20)$$

with closed-loop characteristic polynomial $\mathcal{D}(s) = s^2 + \sqrt{2}\kappa s + \kappa^2$: natural frequency $\omega_n = \kappa$ and damping ratio $\zeta = 1/\sqrt{2}$. The gains do not depend on the noise intensity σ_w^2 .

Proof. Seek a value function $V(x) = x^\top P x$ and average cost ϱ (the average-cost constant; unrelated to the recovery time ρ) satisfying the average-cost Hamilton–Jacobi–Bellman equation

$$\varrho = \min_v \left[qe^2 + rv^2 + \nabla V^\top (Ax + Bv) \right] + \frac{1}{2}\sigma_w^2 \partial_{ee}^2 V.$$

(The verification theorem for average-cost linear-quadratic control is imported; the computation below exhibits the solution it certifies.) The minimization over v is a quadratic: $\partial_v[rv^2 + 2x^\top P B v] = 0$ gives $v^* = -\frac{1}{r}B^\top P x$. Substituting back, the x -dependent part must vanish for all x , giving the algebraic Riccati equation

$$A^\top P + PA - \frac{1}{r}PBB^\top P + Q = 0, \quad Q = \begin{pmatrix} q & 0 \\ 0 & 0 \end{pmatrix},$$

and $\varrho = \sigma_w^2 P_{11}$. Write $P = \begin{pmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{pmatrix}$. Componentwise:

$$A^\top P + PA = \begin{pmatrix} 0 & -p_{11} \\ -p_{11} & -2p_{12} \end{pmatrix}, \quad \frac{1}{r}PBB^\top P = \frac{1}{r} \begin{pmatrix} p_{12}^2 & p_{12}p_{22} \\ p_{12}p_{22} & p_{22}^2 \end{pmatrix}.$$

The three scalar equations are

$$(1,1) : q = \frac{p_{12}^2}{r}, \quad (1,2) : p_{11} = -\frac{p_{12}p_{22}}{r}, \quad (2,2) : -2p_{12} = \frac{p_{22}^2}{r}.$$

Equation (2,2) forces $p_{12} < 0$, so (1,1) gives $p_{12} = -\sqrt{qr}$; then (2,2) gives $p_{22} = \sqrt{2r\sqrt{qr}} = \sqrt{2}q^{1/4}r^{3/4}$ and (1,2) gives $p_{11} = \sqrt{2}q^{3/4}r^{1/4} > 0$. Positive definiteness: $p_{11} > 0$ and $\det P = p_{11}p_{22} - p_{12}^2 = 2qr - qr = qr > 0$. The law is $v = -\frac{1}{r}(p_{12}e + p_{22}u) = \sqrt{q/r}e - \sqrt{2}(q/r)^{1/4}u$, which is eq. (20) with $\kappa = (q/r)^{1/4}$. The closed-loop matrix $A - \frac{1}{r}BB^\top P = \begin{pmatrix} 0 & -1 \\ \kappa^2 & -\sqrt{2}\kappa \end{pmatrix}$ has characteristic polynomial $s(s + \sqrt{2}\kappa) + \kappa^2 = s^2 + \sqrt{2}\kappa s + \kappa^2$, i.e., $\omega_n = \kappa$, $\zeta = \sqrt{2}\kappa/(2\kappa) = 1/\sqrt{2}$, stable. The noise intensity appears only in ϱ , not in the Riccati equation, so the gains are noise-independent. $\square$

Remark 7.2 (Correspondence with the deployed controller). The implementation runs exactly this second-order law (plus integral action, section 7.6), derived earlier in sample units with cost $\text{Var}[e_{\text{samples}}] + \lambda \text{Var}[\dot{u}]$ (that document’s smoothness weight λ , unrelated to the episode rates

λ_i). Converting $e_{\text{samples}} = F_s e_{\text{seconds}}$ maps that cost to $q = F_s^2$, $r = \lambda$, hence $\kappa = (F_s^2/\lambda)^{1/4}$ — the same natural frequency, and the same universal damping $\zeta = 1/\sqrt{2}$. What is new here is not the law but its *scope*: theorem 7.3 makes it the full-information optimum over all causal laws for the difference coordinates — not a choice within a two-parameter family — and S10 states exactly what running it on the raw measurement gives up.

Proposition 7.4 (Stationary variances of the fast loop). *Under the law eq. (20) driven by white noise of intensity σ_w^2 on e , the stationary covariance $X = \text{Cov}(e, u)$ solves the Lyapunov equation $A_c X + X A_c^\top + \text{diag}(\sigma_w^2, 0) = 0$ with $A_c = \begin{pmatrix} 0 & -1 \\ \kappa^2 & -\sqrt{2}\kappa \end{pmatrix}$, and*

$$\text{Var}[e] = \frac{3\sigma_w^2}{2\sqrt{2}\kappa}, \quad \text{Cov}[e, u] = \frac{\sigma_w^2}{2}, \quad \text{Var}[u] = \frac{\kappa\sigma_w^2}{2\sqrt{2}}, \quad \text{Var}[\dot{u}] = \frac{\kappa^3\sigma_w^2}{2\sqrt{2}}. \quad (21)$$

Proof. Write $X = \begin{pmatrix} x_e & x_c \\ x_c & x_u \end{pmatrix}$. Compute $A_c X = \begin{pmatrix} -x_c & -x_u \\ \kappa^2 x_e - \sqrt{2}\kappa x_c & \kappa^2 x_c - \sqrt{2}\kappa x_u \end{pmatrix}$. The Lyapunov equation componentwise:

$$(1,1) : -2x_c + \sigma_w^2 = 0 \Rightarrow x_c = \frac{\sigma_w^2}{2}; \quad (2,2) : 2\kappa^2 x_c - 2\sqrt{2}\kappa x_u = 0 \Rightarrow x_u = \frac{\kappa x_c}{\sqrt{2}} = \frac{\kappa\sigma_w^2}{2\sqrt{2}};$$

$$(1,2) : -x_u + \kappa^2 x_e - \sqrt{2}\kappa x_c = 0 \Rightarrow x_e = \frac{x_u + \sqrt{2}\kappa x_c}{\kappa^2} = \frac{\sigma_w^2}{\kappa} \left(\frac{1}{2\sqrt{2}} + \frac{\sqrt{2}}{2} \right) = \frac{3\sigma_w^2}{2\sqrt{2}\kappa}.$$

For $\text{Var}[\dot{u}]$, use $\dot{u} = v = \kappa^2 e - \sqrt{2}\kappa u$:

$$\text{Var}[v] = \kappa^4 x_e - 2\sqrt{2}\kappa^3 x_c + 2\kappa^2 x_u = \kappa^3 \sigma_w^2 \left(\frac{3}{2\sqrt{2}} - \sqrt{2} + \frac{1}{\sqrt{2}} \right) = \frac{\kappa^3 \sigma_w^2}{2\sqrt{2}},$$

using $\frac{3}{2\sqrt{2}} - \sqrt{2} + \frac{1}{\sqrt{2}} = \frac{3-4+2}{2\sqrt{2}} = \frac{1}{2\sqrt{2}}$. □

7.4 Speaker-space diagonalization

Proposition 7.5 (Mode decomposition). *Let $\Pi = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ be the centering matrix. Then:*

- (i) *The synchronization cost is $\frac{q}{N}L^\top \Pi L$; the matrix Π has eigenvalue 0 on the all-ones direction (the mean coordinate) and eigenvalue 1 on its orthogonal complement (the difference coordinates).*
- (ii) *In any orthonormal basis whose first vector is $\mathbf{1}/\sqrt{N}$, the quadratic part of the loss and the dynamics theorem 6.2 split into: a mean-coordinate problem with zero quadratic state cost, linear cost $\alpha\bar{L}$, and warp-rate cost; and $N-1$ identical difference problems, each an instance of eq. (19) with weights $(q/N, r/N)$ — the common factor $1/N$ cancels in the ratio q/r , so the bandwidth κ of theorem 7.3 is unaffected and the factor is suppressed from here on.*

Proof. (i) $\Pi\mathbf{1} = \mathbf{1} - \mathbf{1} = 0$; for $x \perp \mathbf{1}$, $\Pi x = x$. Also $\sum_i (L_i - \bar{L})^2 = L^\top \Pi L$ by expanding the square. (ii) The warp cost $\frac{r}{N} \sum \dot{u}_i^2 = \frac{r}{N} \dot{u}^\top I \dot{u}$ is proportional to the identity, hence diagonal in every orthonormal basis; the dynamics are the same scalar integrator in every coordinate (theorem 6.2); the synchronization cost is diagonal by (i) with the stated eigenvalues; the latency cost $\alpha\bar{L}$ involves only the first basis vector. Disturbance covariances need not be diagonal in this basis; by theorem 7.3 the optimal gains do not depend on them, and cost evaluation projects the covariances onto the modes. □

Remark 7.3 (What the mean coordinate is left with). The mean coordinate has no quadratic restoring cost: synchronization puts eigenvalue zero on it, and the warp-rate cost charges only *motion* of the mean warp, not its value. Its only costs are the linear latency price $\alpha\bar{L}$ and

(through the margin) the indicator term. The quadratic machinery therefore has nothing to say about *where* the fleet sits — only about how it moves. This degeneracy is not an accident; it is the structural hook on which section 9 hangs the underrun term.

7.5 Closed-loop transfer functions and the component ledger

The fast law eq. (20) in transfer form: from $sU = \kappa^2 Y - \sqrt{2}\kappa U$,

$$U(s) = C(s) Y(s), \quad C(s) = \frac{\kappa^2}{s + \sqrt{2}\kappa}, \quad (22)$$

where Y is the measured error. The plant is $sE = W - U$ with W the drift disturbance. Let $\tilde{D}$ denote the transform of the deviation of the delay process from its baseline — the same physical quantity written x_i in the time domain (section 8.1) — N_i the transform of the measurement error n_i of Definition 2.7, and $\mathcal{D}(s) = s^2 + \sqrt{2}\kappa s + \kappa^2$.

Proposition 7.6 (Backend transfer functions). *Under the margin measurement ($Y = E - \tilde{D}$, Definition 2.8):*

$$\frac{E}{\tilde{D}} = \frac{\kappa^2}{\mathcal{D}(s)}, \quad \frac{M}{\tilde{D}} = -\frac{s(s + \sqrt{2}\kappa)}{\mathcal{D}(s)}, \quad \frac{U}{\tilde{D}} = -\frac{\kappa^2 s}{\mathcal{D}(s)}, \quad (23)$$

where $M = E - \tilde{D}$ is the margin deviation. Under the latency measurement ($Y = E + N_i$, Definition 2.7):

$$\frac{E}{N_i} = -\frac{\kappa^2}{\mathcal{D}(s)}, \quad \frac{E}{\tilde{D}} = 0, \quad \frac{M}{\tilde{D}} = -1 \quad (\text{all frequencies}). \quad (24)$$

Proof. Margin measurement: $sE = W - C(E - \tilde{D})$ gives $E(s + C) = W + C\tilde{D}$, so $E/\tilde{D} = C/(s + C)$. Compute $s + C = (s(s + \sqrt{2}\kappa) + \kappa^2)/(s + \sqrt{2}\kappa) = \mathcal{D}(s)/(s + \sqrt{2}\kappa)$, hence $C/(s + C) = \kappa^2/\mathcal{D}(s)$. Then $M/\tilde{D} = E/\tilde{D} - 1 = (\kappa^2 - \mathcal{D}(s))/\mathcal{D}(s) = -s(s + \sqrt{2}\kappa)/\mathcal{D}(s)$, and $U = C(E - \tilde{D}) = C(M/\tilde{D})\tilde{D} = -\kappa^2 s/\mathcal{D}(s)\tilde{D}$. Latency measurement: $sE = W - C(E + N_i)$ gives $E/N_i = -C/(s + C) = -\kappa^2/\mathcal{D}(s)$; the delay deviation does not enter the loop equations at all, so $E/\tilde{D} = 0$ and $M = E - \tilde{D}$ carries $\tilde{D}$ with gain -1 at every frequency. $\square$

Remark 7.4 (Reading theorem 7.6). Under the margin measurement, $E/\tilde{D}$ is a unit-gain low-pass: slow delay changes (baseline) are converted one-for-one into latency changes, and only fluctuations above the loop bandwidth κ land in the margin. The pause component is far above κ in its onset, so a pause drives the margin almost fully — but the loop still responds through $U/\tilde{D}$: pauses leak into warp (and hence into the synchronization and roughness terms) under this backend. Under the latency measurement the network deviation never drives the loop: the entire deviation, at *all* frequencies, lands in the margin. Slow baseline growth then consumes margin silently — the level layer must track the baseline explicitly (section 9.3) — and a drifting clock error n_i moves true latency one-for-one at DC. Neither backend removes a disturbance; each chooses which loss term absorbs it.

7.6 Integral action

The drift w_i is unknown and slowly varying. The principled treatment augments the state with the unknown input ($\dot{w} = 0$) and solves the augmented problem; by the internal-model principle (imported: Francis and Wonham, 1976), any law achieving zero steady-state error against a constant disturbance must contain an integrator, and the certainty-equivalent reduction of the

Table 1: Which loss term absorbs which disturbance component.

Component	Latency measurement (e2e)	Margin measurement (niq)
Baseline B_i	indicator (silent margin drain; level layer must track $\hat{B}_i$)	latency term (latency follows the network)
Diffusion ε_i	indicator (bulk tail term)	margin (high-pass part) $\rightarrow$ indicator; warp/sync (band-pass leakage)
Pauses P_i	indicator only	indicator + warp/sync leakage
Drift w_i	warp DC (free)	warp DC (free)
Clock error n_i	latency term (unit DC gain)	absent

augmented design — estimate $\hat{w}$ by integrating the loop innovation, feed it forward — lands on the law eq. (20) plus an integral term:

$$\dot{\xi} = e, \quad v = \kappa^2 e + K_0 \xi - \sqrt{2}\kappa u, \quad (25)$$

with K_0 placing a pole well below κ (dominant-pole design). Two caveats, so the status of eq. (25) is exact: it is certainty-equivalent and dominant-pole-tuned, not the jointly optimized augmented-state law (K_0 is set by a chosen settling time T_I , not by the augmented Riccati equation); and the integral mode inflates the stationary error variance by the factor $1/(1 - 1/(\sqrt{2}\kappa T_I)) \approx 1.035$ — a 3.5% overhead at the deployed $\kappa \approx 0.69 \text{ s}^{-1}$, $T_I = 30 \text{ s}$. The deployed controller implements exactly eq. (25) with anti-windup on the warp clamp; the earlier document¹ gives the dominant-pole design, the variance-overhead computation, and stability margins (the integral gain must satisfy $K_0 < \sqrt{2}\kappa \cdot \kappa^2$, the Routh bound).

8 The Indicator Term

The underrun term is outside the quadratic algebra (section 5.2). This section evaluates it by a chain of quantified simplifications. The end product is a function $R_i(\mu)$ — underrun fraction versus margin level — consumed by the level program of section 9.

8.1 Exact rewriting

Fix receiver i . Under S1, the underrun contribution to the loss is c times the stationary probability $\mathbb{P}(m_i < 0)$. Decompose the margin using Assumption 3.1: with the fast loop holding the latency near a slowly varying level $\ell_i(t)$,

$$m_i = \ell_i - B_i - x_i + e_i^{\text{loop}}, \quad x_i := \varepsilon_i + P_i, \quad (26)$$

where x_i is the *deviation* (the part of the delay above baseline; this is the quantity the telemetry’s deviation meter measures) and e_i^{loop} is the fast loop’s tracking error. Define the *margin level*

$$\mu_i := \ell_i - \hat{B}_i \quad (\text{latency backend}), \quad (27)$$

the distance between the held latency and the estimated baseline. Under the margin backend the loop regulates the margin directly and μ_i is the configured queue target itself, with no baseline subtraction (Remark 8.1); every use of μ_i below is backend-conditional in this sense. Then

$$\mathbb{P}(m_i < 0) = \mathbb{P}(x_i > \mu_i) + (\text{errors}), \quad R_i(\mu) := \mathbb{P}(x_i > \mu), \quad (28)$$

¹`controller.tex` in this directory: the second-order controller derivation. Its gain optimization is consistent with theorem 7.3; its noise passage is superseded by theorems 3.2 and 3.3: a stationary jitter level cannot supply the accumulating white noise assumed there, so its σ_w must be read as baseline wander plus measurement noise, and its Gaussian-tail underrun estimates are replaced by section 8.

where the errors come from baseline estimation error $B_i - \hat{B}_i$, bounded by the estimator lag $\bar{\beta} \tau_{\hat{B}}$ (with $\tau_{\hat{B}}$ the baseline filter's time constant, distinct from the emission time $\tau(s)$), and from e_i^{loop} , whose stationary scale is $\sqrt{\text{Var}[e]} = \sqrt{3\sigma_w^2/(2\sqrt{2}\kappa)}$ by theorem 7.4 — both far smaller than σ_i at the deployed constants. R_i is the exceedance function of the stationary deviation law — a directly measurable object: the deviation histogram estimates it without any model.

Remark 8.1 (Backend dependence of the deviation law). Equation (26) reads differently per backend. Under the latency measurement, the loop holds L_i at ℓ_i and the full deviation lands in the margin at every frequency (theorem 7.6): x_i enters unfiltered. Under the margin measurement, the loop absorbs the part of x_i below the bandwidth κ into latency, so the deviation reaching the margin is high-passed, and the margin level μ_i is the configured queue target itself rather than $\ell_i - \hat{B}_i$ (the baseline is absorbed by the loop and drops out of eq. (27)). In both cases the deviation histogram is measured on the running closed loop, so $\hat{R}_i$ estimates the correct backend-specific exceedance law by construction; only the component-model *extrapolation* of section 8.2 must respect the backend.

Definition 8.1 (Occupation density). $g_i(\mu) := -dR_i/d\mu$: the probability density of time the deviation spends at level μ . The level program's optimality condition (section 9.3) is stated in terms of g_i .

8.2 Evaluation under the component model

The component model of section 3 evaluates R_i as three terms. The decomposition serves two purposes: it *extrapolates* R_i beyond the directly observed range (the tail of the pause height law extends past any finite observation window), and it *explains* which mechanism dominates at which margin.

Proposition 8.1 (Risk evaluation). *Under S1–S5 below,*

$$R_i(\mu) = \underbrace{\mathbb{P}(\varepsilon_i > \mu)}_{\text{bulk}} + \underbrace{\lambda_i \mathbb{E}(H_i - \mu)^+}_{\text{pauses (theorem 5.5)}} + \underbrace{\lambda_i \int \mathbb{E}\left[(H' - \mu + \delta(g))^+ - (H' - \mu)^+\right] G_i(dg)}_{\text{compounding}} + \text{defects of S2–S5}, \quad (29)$$

where G_i is the gap law between consecutive episodes, H' is the height of the following episode, and

$$\delta(g) = \chi (1 - g/\rho)^+ \quad (30)$$

is the residual margin depression at gap g after an episode that consumed margin $\chi = \min(H, \mu)$ and recovers over time ρ (Assumption 3.5; linear recovery model). The expectation in the compounding term is over three quantities: the predecessor's height H and recovery ρ (entering through δ), and the follower's height H' ; per S5 they are taken mutually independent and independent of the gap g , each with its marginal law.

The bulk term is the stationary exceedance of the diffusion — exact under S1 outside episodes. **With wired-host σ_i up to one millisecond and margin levels of order 5–10 ms, the bulk term is comparable to or larger than the pause term;** no negligibility is claimed anywhere. The pause term is theorem 5.5. The compounding term prices an episode that arrives while the margin is still depressed: the follower's effective margin is $\mu - \delta(g)$ instead of μ . With fast redelivery ($\rho \sim$ milliseconds) the integral is negligible because $G_i(\rho) \approx 0$; with warp-limited recovery (ρ up to $\chi/u_{\max}$, i.e., seconds) and the measured gap clustering (short gaps two to three orders above the exponential prediction), the term is first-order. It is computable from measured objects: the gap histogram, the height law, and the recovery mechanism.

Definition 8.2 (Belief state). The *belief state* $\hat{\theta}(t)$ is the vector of estimator outputs feeding the

risk evaluation and, later, the level program:

$$\hat{\theta} := (\hat{B}_i, \hat{\sigma}_i, \text{deviation histogram}, \hat{\lambda}_i, \hat{F}_i, \hat{G}_i, \hat{\rho}_i)_{i=1,\dots,N}.$$

Wherever a later statement says “the belief moves,” it means these estimates change.

8.3 The simplification chain

Simplification 2 (S2: additive component evaluation). **Statement.** R_i is evaluated as the sum eq. (29): bulk and pause exceedances add, and the diffusion is ignored *inside* episodes.

Role. Lets each term be estimated by its own estimator (the histogram for the bulk; the event meter for pauses).

Defect. Three parts. (a) *Threshold smearing*: inside an episode the true condition is $\varepsilon_i + P_i > \mu$, not $P_i > \mu$; because the ramp crosses level μ at unit rate, the diffusion shifts the crossing time by $O(|\varepsilon_i|)$, so the error per episode is at most $\mathbb{E}|\varepsilon_i|$ seconds of misclassified time, and the rate error is at most $\lambda_i \bar{F}_i(\mu - O(\sigma_i)) \cdot O(\sigma_i)$ relative to a pause term whose scale is $\lambda_i \mathbb{E}(H_i - \mu)^+$: the relative defect is $O(\sigma_i/\text{height spread})$, a few percent when heights are tens of milliseconds, tens of percent when the height law concentrates near the margin — in which case the bulk and pause terms have merged physically and the histogram estimate of R_i (which never used the decomposition) remains the ground truth. (b) *Duty cycle*: the bulk term uses all time rather than time outside episodes; the error is a factor $1 - \lambda_i \mathbb{E}[H_i]$ — order 10^{-3} on wired hosts, up to order 10^{-2} on radio hosts with second-scale heights. (c) *Zero-drain ramp* (theorem 3.1): per-episode timing error bounded by the burst drain time of Assumption 3.5(a).

Monitor. Compare the composed R_i of eq. (29) against the raw deviation-histogram exceedance on the overlap range; disagreement is exactly the S2 defect.

Simplification 3 (S3: the fast layer cannot move the indicator). **Statement.** The underrun term is treated as independent of the fast-loop policy: only the level path $\ell_i(t)$ enters it.

Role. This is the load-bearing decoupling. It removes the indicator from the fast layer’s optimization, which is what allows theorem 7.3 to stand as the fast-layer optimum. Without it, the barrier term couples to the loop and the problem is a free-boundary dynamic program.

Defect. During an episode, the most any admissible policy can add to the margin before it empties is $u_{\max} \times (\text{time to empty}) = u_{\max} \mu$ (playing slower stretches μ buffered seconds into $\mu/(1 - u_{\max})$ wall seconds). With $u_{\max} = 5 \times 10^{-3}$ and $\mu = 10$ ms this is $50 \mu\text{s}$; with $\mu = 100$ ms, 0.5 ms. The loss difference between the best fast policy and a fast policy that ignores the indicator is therefore at most

$$\frac{c}{N} \sum_i [R_i(\mu_i) - R_i(\mu_i(1+u_{\max}))] \leq \frac{c}{N} \sum_i g_i(\mu_i) u_{\max} \mu_i,$$

a relative defect of $u_{\max} \mu_i g_i(\mu_i)/R_i(\mu_i)$. The honest size of this ratio differs by component. For the pause part, g/R is of order one over the height spread, so the ratio is $u_{\max} \mu/(\text{height spread})$ — well below one percent at the reference numbers. For the bulk part, g/R is the tail hazard rate of ε_i , of order $1/x_0$ with $x_0 \leq \sigma_i$ the tail scale: the ratio is $u_{\max} \mu/x_0$, which at $\mu = 10$ ms and $x_0 = 1$ ms is 5%, and larger for faster-decaying tails. The decoupling nevertheless stands, on an absolute anchor: at the operating point the first-order condition (theorem 9.2) pins $\frac{c}{N} \sum_i g_i(\mu_i) = \alpha$, so the absolute loss the fast layer could recover is at most $c \bar{g} u_{\max} \mu = \alpha u_{\max} \mu$ — a fraction $u_{\max} \mu/\ell \leq u_{\max} = 0.5\%$ of the latency term itself. No admissible fast policy moves the indicator term by more than that. The same authority argument covers reacting to bulk fluctuations: a fluctuation of duration $T_f \sim 10$ ms grants $u_{\max} T_f = 50 \mu\text{s} \ll \sigma_i$.

Monitor. Both forms: the relative ratio $u_{\max} \mu g(\mu)/R(\mu)$ per component, and the absolute anchor $\alpha u_{\max} \mu$ against the running loss estimate; if warp authority is ever raised substantially, these monitors flag the breakdown of the decomposition (section 12).

Simplification 4 (S4: frozen margin during an episode). **Statement.** The evaluation eq. (16) holds the margin constant during an episode: no mid-episode warp response and no mid-episode refill.

Role. Gives the closed form $(H - \mu)^+$ for per-episode underrun time.

Defect. Mid-episode warp is bounded by S3 ($\leq u_{\max}\mu$); mid-episode arrivals are excluded by the definition of an episode (delivery stopped). The defect is the S3 number again; it is not counted twice in the register.

Monitor. Same as S3.

Simplification 5 (S5: one-step compounding). **Statement.** The compounding term in eq. (29) considers only consecutive episode pairs, with the follower’s height drawn independently of the gap; chains of three or more episodes inside one recovery window are dropped, as is gap–height dependence.

Role. Keeps the compound term a one-dimensional integral over the measured gap histogram.

Defect. Chain terms are $O((\lambda_i^{\text{loc}}\rho)^2)$ where λ_i^{loc} is the *local* episode rate inside a cluster (the measured short-gap rate, not the global average) — with strong clustering this is the first quantity to watch; gap–height dependence is measurable from the joint (gap, height) records and enters as a covariance correction.

Monitor. Fraction of episodes whose predecessor lies within ρ ; the drain validator (windowed margin minimum), which sees compound drains directly and arbitrates the whole of eq. (29) against reality.

Simplification 6 (S6: two-timescale separation). **Statement.** The level $\ell_i(t)$ and the belief $\hat{\theta}(t)$ move slowly relative to episodes and to the fast-loop bandwidth: the evaluation treats μ_i as constant within episodes and within estimation windows.

Role. Turns the indicator into a *static* function $R_i(\mu_i)$ of the current level; the level path enters only through the glide cost (section 9.5).

Defect. During a level glide at rate γ , the margin during an episode of duration H differs from the static value by up to γH ; relative defect $\gamma H/\sigma$ -scale. With the glide bounded by $u_{\max}$ and $H \leq 1$ s this is at most 5 ms — not negligible during fast raises; the raise policy of section 9.5 therefore counts the transient explicitly rather than relying on S6.

Monitor. Glide rate telemetry; episodes stamped with the concurrent glide rate.

Simplification 7 (S7: certainty equivalence). **Statement.** The level program optimizes against the *estimated* risk functions $\hat{R}_i$ as if they were true; no hedging against estimation error and no active probing.

Role. Replaces a Bayesian dynamic program over beliefs with a plug-in static program.

Defect. Second-order in the estimation error: for the scalar level program with optimum μ^* , the excess loss is $\frac{1}{2}J''(\mu^*)\mathbb{E}[(\hat{\mu}^* - \mu^*)^2]$ with curvature $J''(\mu^*) = -\frac{c}{N}\sum_i g'_i(\mu_i^*)$, positive under S8’s decreasing densities; the estimator variance follows from event-count statistics: with n exceedance observations near the operating point, $\text{sd}(\hat{\mu}^*) \propto 1/\sqrt{n}$ with a clustering inflation factor read from the gap law (Remark 5.3). Active probing is unnecessary because every episode’s height is measured whether or not it causes underrun: the risk curve at levels above the operating point is identified from the same records, without ever running the system at those levels.

Monitor. Confidence band of $\hat{R}_i$ propagated to a confidence interval on μ^* ; the band width is the S7 budget.

Simplification 8 (S8: smooth, convex risk curve). **Statement.** R_i is differentiable at the operating point (g_i exists), and the composed level objective J is convex on the operating range.

Role. Lets the level program use the first-order condition and report two-sided sensitivities.

Defect. Three ways the statement can fail, none of which breaks the program — it is one-dimensional and solved by direct scan of the measured $\hat{J}$ when the assumption fails (theorem 8.2(ii)). (a) An atom in the height law (a mechanism producing a fixed-height pause) makes

R_i kinked: the first-order condition becomes a subgradient inclusion and the optimum may sit exactly at the atom; sensitivities become one-sided. (b) Convexity of the bulk term requires a decreasing density beyond the operating point (true for unimodal light-tailed laws); the pause term is convex unconditionally ($d^2/d\mu^2 \lambda \mathbb{E}(H - \mu)^+ = \lambda f_H(\mu) \geq 0$). (c) The compounding term carries a concave piece proportional to the residual-depression fraction (theorem 8.2(i) and its proof): in the clustered, slow-recovery regime the composed J can lose convexity, and the first-order characterization with it.

Monitor. The estimated $\hat{g}_i$ near the operating point; mass concentration at single heights; the sign of the second difference of the scanned $\hat{J}$ across the operating range.

Simplification 9 (S9: component independence in second-order accounting). **Statement.** Cross-covariances between B_i , ε_i , P_i (and across receivers, between idiosyncratic parts) are dropped where second moments are summed.

Role. Keeps the spectral accounting of section 7.5 per-component.

Defect. Measured violations exist: episodes co-move with baseline shifts (a radio environment change moves both). The dropped terms are covariances bounded by the product of component scales; the affected outputs are the quadratic terms' *evaluations* (not the gains, which are covariance-independent by theorem 7.3, and not the risk curve, which is estimated on the summed deviation directly).

Monitor. Cross-correlation of the baseline estimator's drift with event occurrence.

8.4 Shape of the risk curve

Proposition 8.2 (Monotonicity, convexity, interiority). *R_i is nonincreasing. Consider the composed level objective over the fleet level ℓ , $J(\ell) = \alpha\ell + \frac{c}{N} \sum_i R_i(\ell - b_i)$ (with fixed baseline offsets b_i), on the domain $\ell \geq \ell_{\text{floor}} := \max_i b_i$ (below the floor, some receiver's margin is negative — permanent underrun — and such levels are excluded from the program). Then:*

- (i) *The bulk and pause contributions to each R_i are convex under S8's density conditions; the compounding contribution is not convex in general (see the proof), so convexity of the composed J on the operating range is an assumption, carried by S8 and monitored, not a theorem.*
- (ii) *J is continuous and $J(\ell) \geq \alpha\ell \rightarrow \infty$, so a minimizer exists on $[\ell_{\text{floor}}, \infty)$ regardless of convexity; because the program is one-dimensional over a measured function, the minimizer is computable by direct scan of $\hat{J}$, and eq. (33) characterizes it only where J is smooth and locally convex.*
- (iii) *Under S8 (including composed convexity), the minimizer is unique and is interior — so that the first-order condition of theorem 9.2 applies — exactly when*

$$\alpha < \frac{c}{N} \sum_{i=1}^N g_i(\ell_{\text{floor}} - b_i), \quad (31)$$

the risk relief available at the floor (each receiver evaluated at its own floor margin $\ell_{\text{floor}} - b_i \geq 0$); otherwise the optimum sits at the floor — a corner worth an alarm, not a controller.

Proof. Monotonicity: exceedance functions are nonincreasing. Convexity of the bulk part where the density is decreasing; of the pause part because $d^2/d\mu^2 \mathbb{E}(H - \mu)^+ = f_H(\mu) \geq 0$ by differentiating $\mathbb{E}(H - \mu)^+ = \int_{\mu}^{\infty} \bar{F}(h) dh$ twice. The compounding part fails the same argument: its threshold for the follower is $t(\mu) = \mu - c_g \min(H, \mu)$ with $c_g = (1 - g/\rho)^+$ the residual-depression fraction, so $t'(\mu)$ jumps up by c_g at $\mu = H$ (from $1 - c_g$ to 1), the slope $-t'(\mu)\bar{F}_{H'}(t(\mu))$ jumps down, and mixing over the predecessor height contributes the negative (concave) piece $-c_g \bar{F}_{H'}((1 - c_g)\mu) f_H(\mu)$ to J'' alongside the positive $t'^2 f_{H'}$ pieces. The concave piece scales with

c_g : it is largest exactly in the clustered, slow-recovery regime. Hence (i). Existence by continuity and coercivity upward; the scan statement is immediate from one-dimensionality. Interiority: $J'(\ell_{\text{floor}}^+) = \alpha - \frac{c}{N} \sum_i g_i(\ell_{\text{floor}} - b_i)$; under composed convexity, eq. (31) makes this negative and the minimizer interior; if it fails, $J' \geq 0$ on the whole domain and the floor is optimal. $\square$

9 Composition: the Two Solvers and the Free Direction

The quadratic machinery (section 7) and the risk evaluation (section 8) use disjoint mathematics and disjoint statistics. This section shows why they compose without conflict, derives the level program, and states the decomposition theorem.

9.1 The free direction

Collect three facts already proven:

- (F1) The synchronization cost has eigenvalue zero on the mean coordinate (theorem 7.5): moving all receivers together is free for synchronization.
- (F2) The warp cost charges only the *rate of change* of warp: a held latency level costs nothing, and a steady glide of the level costs nothing either — during a constant-rate glide the commanded mean warp is constant, so its contribution to $\dot{u}$ is zero. Only changes of the glide rate are charged, through $\dot{u} = -\dot{\ell}$ up to two residuals — drift-rate variation and the common-mode part of the fast loops’ disturbance response — both priced by S13 below (theorems 6.1 and 6.2).
- (F3) After S2–S6, the indicator term depends on the policy only through the slowly varying margin levels $\mu_i(t) = \ell_i(t) - \hat{B}_i(t)$ (section 8.2).

The quadratic part of the loss is thus *degenerate* exactly on the coordinate — the static mean level — through which the indicator term acts. The two solvers meet on orthogonal coordinates: the quadratic solver fills every mode and every frequency except the mean coordinate’s zero-frequency component; the risk program occupies precisely that component. The loss was shaped for this: the linear latency term (no quadratic pull toward any preset level), the variance-only synchronization term (no absolute reference), and the derivative-only warp cost (no charge for holding an offset) each keep the static mean level free of quadratic forces.

Simplification 13 (S13: common-mode loop response priced separately). **Statement.** The decomposition below identifies the mean coordinate’s warp cost with the level path’s roughness, $\dot{u} = -\dot{\ell}$. Two residual contributions to $\dot{u}$ are set aside and accounted here: (a) the *common-mode disturbance response* — every receiver’s fast loop responds to its own measured disturbance, and the fleet-average of those responses (shared wander, shared measurement noise, co-occurring pauses; Assumption 3.6) is mean warp motion belonging to neither the difference modes nor the level path; (b) *drift-rate variation*, $\dot{w} \neq 0$.

Role. Keeps $J_{\text{diff}} + J_{\text{mean}}$ additive in theorem 9.1.

Defect. (a) is computable from the transfer functions of theorem 7.6 evaluated on the *common* component’s spectrum: the omitted cost is $r \int |sU/Y(j\omega)|^2 S_{\text{common}}(\omega) d\omega/2\pi$ for the relevant backend input, the same integral that prices per-receiver roughness, just fed the common spectrum — the “warp/sync leakage” entries of table 1 name exactly this mechanism. Under the latency measurement on disciplined hosts the common driver is small; under the margin measurement co-occurring pauses make it the dominant residual. (b) is bounded by clock physics: quartz frequency drift is parts in 10^{10} per second (thermal); taking a two-orders-of-magnitude margin, $|\dot{w}_i| \leq 10^{-8} \text{ s}^{-1}$, the contribution $r \mathbb{E}[\dot{w}^2] \leq 10^{-16} r$ is negligible at any sane r .

Monitor. Cross-receiver coherence of the loop innovations (estimates S_{common}); the fleet-average warp-rate spectrum against the level layer’s commanded $\dot{\ell}$.

9.2 The decomposition theorem

Definition 9.1 (Layered policies). A policy is *layered* when it consists of: a level path $\ell(t)$, adapted to the sender’s information and broadcast to the receivers; and per-receiver fast laws that act causally on the receiver’s own measured error relative to the broadcast level. Write $\mathcal{P}_L$ for this class. A layered policy is specified by the pair $(\pi_{\text{fast}}, \ell(\cdot))$.

Theorem 9.1 (Structure of the optimal controller). *Under S1–S13:*

(0) **Decomposition.** For every layered policy $\pi = (\pi_{\text{fast}}, \ell(\cdot)) \in \mathcal{P}_L$,

$$J(\pi) = \underbrace{J_{\text{diff}}(\pi_{\text{fast}})}_{\text{difference modes}} + \underbrace{J_{\text{mean}}(\ell(\cdot))}_{\text{level path}} + \Delta_S(\pi),$$

where the evaluation error $\Delta_S(\pi)$, in loss-rate units (u/s), is bounded by the sum of the S1–S13 defects, each converted to an absolute loss-rate contribution (relative defects multiply the term they qualify; the register of section 11 lists both forms).

Within $\mathcal{P}_L$:

(i) J_{diff} depends only on the fast-layer feedback and is minimized, independently of the level path and within the policy class of S10, by the law $v = \kappa^2 e - \sqrt{2}\kappa u$ of theorem 7.3 applied in each difference coordinate (equivalently: by every receiver running that law against the common broadcast level), with $\kappa = (q/r)^{1/4}$. Under exact error observation the class restriction is vacuous and the law is optimal among all causal laws (theorem 7.3 and Remark 7.1).

(ii) J_{mean} depends only on the level path:

$$J_{\text{mean}} = \mathbb{E} \left[\alpha \ell + \frac{c}{N} \sum_{i=1}^N R_i(\ell - \hat{B}_i) + r \left(\frac{d^2 \ell}{dt^2} \right)^2 \right] \quad (32)$$

under the stationary law of the belief process, and is minimized within the hold-and-glide subclass — hold ℓ at the argmin of the static program (section 9.3); rest-to-rest glides per section 9.5 when the argmin moves — up to the S6 and S7 defects. Anticipatory level policies, which act on forecasts of the belief rather than its current value (the direction endorsed by theorem 9.5), are not derived here.

Proof. Assembly of proven pieces. By theorem 6.1 the pointwise loss splits into mean terms, difference terms, and the indicator. By theorem 6.2 the coordinates have decoupled dynamics, so a policy may be specified per coordinate; conversely any policy’s loss evaluates coordinate-wise on the split terms. The difference part is the stationary expectation (S1) of the quadratic cost eq. (19) per coordinate, minimized by theorem 7.3 within the policy class of S10; the minimizing gains do not depend on the disturbance covariances, so the same law is optimal in every difference coordinate regardless of cross-receiver correlation. The indicator, after S2 (additive evaluation), S3–S4 (no fast-policy dependence), S5 (compounding closed over the gap law) and S6 (levels static on episode timescales), equals $\frac{c}{N} \sum_i R_i(\mu_i)$ with R_i from theorem 8.1 — a function of the level path alone. The remaining mean-coordinate quadratic cost is $r \mathbb{E}[\dot{u}^2]$; on level glides $\dot{u} = -\ddot{\ell}$ up to the two residuals of S13 (drift-rate variation and the common-mode disturbance response), so its stationary cost is the third term of eq. (32) plus S13’s registered contribution (S6 separates the timescales so no cross-frequency terms survive). The per-receiver level tilts are set to zero by S11 (section 9.4); every dropped term is one of the S-defects, collected in Δ_S . $\square$

Remark 9.1 (The optimality gap of the layered class — an assembled claim, not a theorem). We also claim: the best layered policy is within the same defect budget of the best causal policy. The reasoning: the mechanisms an unrestricted policy could exploit beyond the layered class are

the ones the simplifications discard, and each S-item bounds its mechanism's worth (S3: in-event control; S10: filtering and nonlinear response to the measurement; S11: per-receiver levels; S13: coordinated common-mode response; S7: belief-coupled control). This is an *assembled* bound — each piece rigorous under its own stated conditions, summed — and the completeness of the mechanism list is an engineering judgment, not a proven exhaustion of policy space. It sits outside the theorem for exactly that reason.

9.3 The level program

Proposition 9.2 (Operating point). *The static part of eq. (32) is minimized at the level ℓ^* solving*

$$\frac{c}{N} \sum_{i=1}^N g_i(\ell^* - \hat{B}_i) = \alpha \quad \Longleftrightarrow \quad \frac{1}{N} \sum_{i=1}^N g_i(\mu_i^*) = \frac{\alpha}{c}, \quad (33)$$

with g_i the occupation density (Definition 8.1); under S8 and the interiority condition eq. (31) the minimizer exists, is unique, and satisfies eq. (33) (theorem 8.2). Only the ratio α/c enters: the operating point balances the summed occupation density of the deviation at the margin against the price ratio of latency to underrun. Dimensional check: $[\alpha/c] = s^{-1} = [g_i]$ (Remark 4.1).

Proof. Differentiate $\alpha\ell + \frac{c}{N} \sum_i R_i(\ell - \hat{B}_i)$ in ℓ and use $R'_i = -g_i$; convexity (theorem 8.2) makes the stationary point the minimum. $\square$

Remark 9.2 (The target has emerged). No reference latency appears anywhere in sections 2 to 4. The quantity ℓ^* of eq. (33) is produced by the optimization: it is a function of the estimated baseline levels and risk curves — that is, of the controller's belief state — and it moves when the belief moves. The “latency target” of the implementation is thus a *dynamic state variable of the controller*, with dynamics inherited from the estimators through the argmin map, rate-limited by section 9.5. This is the promised answer to the question the problem statement left open: a reference *may* arise in the solution, and it does — as internal state, not as input.

9.4 Per-receiver tilts

Allowing each receiver its own level ℓ_i turns the static program into $\min_{\ell_1, \dots, \ell_N} \alpha\bar{\ell} + \frac{q}{N} \sum_i (\ell_i - \bar{\ell})^2 + \frac{c}{N} \sum_i R_i(\ell_i - \hat{B}_i)$.

Proposition 9.3 (Optimal tilts). *The stationary conditions are, for each i ,*

$$\frac{\alpha}{N} + \frac{2q}{N} (\ell_i - \bar{\ell}) = \frac{c}{N} g_i(\mu_i),$$

whose average over i recovers eq. (33), and whose centered part gives

$$\ell_i - \bar{\ell} = \frac{c [g_i(\mu_i) - \frac{1}{N} \sum_j g_j(\mu_j)]}{2q}. \quad (34)$$

Each receiver is displaced by its excess risk pressure relative to the fleet average, divided by twice the synchronization stiffness.

Proof. Differentiate in ℓ_i : $\partial_{\ell_i} [\frac{q}{N} \sum_j (\ell_j - \bar{\ell})^2] = \frac{2q}{N} (\ell_i - \bar{\ell})$ (the cross terms cancel because $\sum_j (\ell_j - \bar{\ell})$ is invariant to ℓ_i shifts through $\bar{\ell}$; expanding, $\partial_{\ell_i} \sum_j (\ell_j - \bar{\ell})^2 = 2(\ell_i - \bar{\ell})(1 - \frac{1}{N}) - \frac{2}{N} \sum_{j \neq i} (\ell_j - \bar{\ell}) = 2(\ell_i - \bar{\ell})$). The latency term contributes α/N and the risk term $-\frac{c}{N} g_i$. Averaging the conditions eliminates the centered term; subtracting the average isolates eq. (34). $\square$

Simplification 11 (S11: tilts set to zero). **Statement.** The broadcast level is common: $\ell_i \equiv \ell^*$.

Role. One number to distribute; receivers need no per-receiver risk data.

Defect. By eq. (34) the forfeited displacement is of order $c \Delta g / (2q)$ with Δg the cross-receiver spread of risk pressure; at the operating point $c g_i \sim \alpha$, so the tilt scale is $\alpha / (2q)$ — with any synchronization weight strong enough to hold sub-millisecond skew, microseconds. The forfeited loss is quadratic in the tilt: with every tilt at its scale, $\frac{q}{N} \sum_i (\text{tilt}_i)^2 \sim \alpha^2 / (4q)$, negligible against every other defect.

Monitor. The spread of estimated $c g_i(\mu_i)$ across receivers, reported next to α .

9.5 Moving the level: glides and asymmetry

The argmin $\ell^*(\hat{\theta})$ moves when the belief moves. Moving the fleet level at rate γ means every receiver warps by γ : the glide is executed by the existing fast loops tracking the moving broadcast and is bounded by $|\gamma| \leq u_{\max}$. Under the loss, a *steady* glide is free — the warp term charges $\dot{u}^2 = \ddot{\ell}^2$, and a constant-rate glide has $\ddot{\ell} = 0$. What a level move costs is therefore: the time-integral of the latency overhang (descents) or of the excess risk (ascents), plus the roughness of *changing* the glide rate. The descent problem has a closed-form solution.

Proposition 9.4 (Optimal descent). *Suppose the belief lowers the argmin by Δ : the current level sits Δ above the target. Write the level relative to the target, $\phi(t) := \ell(t) - \ell^*$, and consider rest-to-rest descents: $\phi(0) = \Delta$, $\dot{\phi}(0) = 0$, $\phi(T) = 0$, $\dot{\phi}(T) = 0$, with the duration T free, subject to no undershoot ($\phi \geq 0$; below the target the risk term rules, so overshoot credit is not available). The cost of a descent path is*

$$C[\phi] = \int_0^T \left[\alpha \phi(t) + r \ddot{\phi}(t)^2 \right] dt \quad (35)$$

(excess latency plus warp roughness; risk never enters because $\phi \geq 0$ keeps the margin at or above its target value throughout, and R_i is nonincreasing). The optimal descent is the quartic arc

$$\phi(t) = \Delta \left(1 - \frac{t}{T^*}\right)^3 \left(1 + \frac{3t}{T^*}\right), \quad \boxed{T^* = 2\sqrt{3} \left(\frac{r\Delta}{\alpha}\right)^{1/4}} \quad (36)$$

with monotone level, peak descent rate and total cost

$$\gamma_{\text{pk}} = \frac{8\sqrt{3}}{27} \Delta^{3/4} \left(\frac{\alpha}{r}\right)^{1/4} \quad \text{at } t = T^*/3, \quad C^* = \frac{16\sqrt{3}}{15} \Delta^{5/4} \alpha^{3/4} r^{1/4}, \quad (37)$$

the cost splitting 3:1 between latency overhang and warp roughness. This holds when $\gamma_{\text{pk}} \leq u_{\max}$, i.e., for $\Delta \leq \Delta_{\text{sat}} := (27u_{\max}/(8\sqrt{3}))^{4/3} (r/\alpha)^{1/3}$; the saturated case follows the proposition.

Proof. Euler–Lagrange. The integrand $F(\phi, \ddot{\phi}) = \alpha \phi + r \ddot{\phi}^2$ has $\partial F / \partial \phi = \alpha$, $\partial F / \partial \dot{\phi} = 0$, $\partial F / \partial \ddot{\phi} = 2r \ddot{\phi}$. For a second-order Lagrangian with ϕ and $\dot{\phi}$ fixed at both ends, the Euler–Lagrange equation is $\partial_\phi F - \frac{d}{dt} \partial_{\dot{\phi}} F + \frac{d^2}{dt^2} \partial_{\ddot{\phi}} F = 0$, i.e., $\alpha + 2r \phi^{(4)} = 0$: constant fourth derivative $\phi^{(4)} = -a$ with $a := \alpha / (2r)$.

The quartic family. Integrating and imposing $\phi(0) = \Delta$, $\dot{\phi}(0) = 0$, $\phi(T) = 0$, $\dot{\phi}(T) = 0$:

$$\phi(t) = -\frac{at^4}{24} + \frac{c_3 t^3}{6} + \frac{c_2 t^2}{2} + \Delta, \quad c_3 = \frac{aT}{2} + \frac{12\Delta}{T^3}, \quad c_2 = -\frac{aT^2}{12} - \frac{6\Delta}{T^2}.$$

For each fixed T the functional $C[\phi]$ is strictly convex in ϕ (a positive quadratic form in $\ddot{\phi}$ plus a linear term), so the Euler–Lagrange solution is the unique global minimizer among all paths with these boundary values — comparing family members across T therefore compares true

per- T optima. Substituting and integrating termwise gives the cost of the family member with duration T :

$$C(T) = \frac{12\Delta^2 r}{T^3} + \frac{\alpha\Delta T}{2} - \frac{\alpha^2 T^5}{2880r}. \quad (38)$$

No interior minimum; the boundary of admissibility. Differentiate and substitute $x = T^4$: $C'(T) = \frac{\alpha\Delta}{2} - \frac{36\Delta^2 r}{x} - \frac{\alpha^2 x}{576r}$. As a function of $x > 0$ this is concave with maximum at $x = 144\Delta r/\alpha$ (set the derivative $36\Delta^2 r/x^2 - \alpha^2/(576r)$ to zero), where its value is $\frac{\alpha\Delta}{2} - \frac{\alpha\Delta}{4} - \frac{\alpha\Delta}{4} = 0$. Hence $C'(T) \leq 0$ for all T , with equality exactly at $T^4 = 144\Delta r/\alpha$, i.e., at $T = T^*$ of eq. (36): the cost decreases in T everywhere else. The minimizer must therefore sit at the largest admissible T .

Admissibility ends at T^ .* Factor the velocity: $\dot{\phi}(t) = -\frac{a}{6}t(t-T)(t-t_1)$ with $t_1 = \frac{T}{2} + \frac{36\Delta}{aT^3}$ (the cubic $\dot{\phi}$ has roots at 0 and T by construction; the product and sum of the quadratic factor's roots give t_1). Now $t_1 \geq T \iff 72\Delta \geq aT^4 \iff T \leq T^*$. For $T \leq T^*$: on $(0, T)$ the factors give $t > 0$, $(t-T) < 0$, $(t-t_1) < 0$, so $\dot{\phi} < 0$: the descent is monotone and $\phi \geq 0$ holds. For $T > T^*$: $t_1 < T$, so $\dot{\phi} > 0$ on (t_1, T) ; with $\phi(T) = 0$ this means $\phi < 0$ just before T — the interior path undershoots and is inadmissible. One admissible alternative remains at long horizons: touch $\phi = 0$ with $\dot{\phi} = 0$ at some $T' < T$ and rest there. Resting at the target costs nothing, so such a path is exactly a descent of duration T' ; admissibility forces $T' \leq T^*$, and C decreasing on $(0, T^*]$ gives $C(T') \geq C(T^*)$. Hence the constrained optimum is exactly $T = T^*$.

Values at T^ .* At $T = T^*$, $t_1 = T^*$ and $\dot{\phi} = -\frac{a}{6}t(t-T^*)^2$: the arrival is tangential. Integrating gives the closed form eq. (36) (check: $\phi(0) = \Delta$, and expanding $\Delta(1-t/T^*)^3(1+3t/T^*)$ reproduces the quartic with $\phi^{(4)} = -24\Delta \cdot 3/T^{*4} = -72\Delta/T^{*4} = -a$ by $T^{*4} = 144\Delta r/\alpha$ and $a = \alpha/2r$ ✓). The peak rate: $\frac{d}{dt}[t(t-T^*)^2] = (t-T^*)(3t-T^*)$ vanishes at $t = T^*/3$, where $\dot{\phi} = -\frac{a}{6} \cdot \frac{T^*}{3} \cdot \frac{4T^{*2}}{9} = -\frac{2aT^{*3}}{81}$; substituting a and T^* gives γ_{pk} of eq. (37). The cost: substituting T^* into eq. (38) gives $C^* = \frac{16\sqrt{3}}{15}\Delta^{5/4}\alpha^{3/4}r^{1/4}$, of which the latency part is $\frac{4\sqrt{3}}{5}(\cdot)$ and the roughness part $\frac{4\sqrt{3}}{15}(\cdot)$: ratio 3:1. $\square$

Remark 9.3 (Saturated descents). For $\Delta > \Delta_{\text{sat}}$ the arc's peak rate would exceed the clamp. The constrained-optimal profile then cruises at u_{max} with entry and exit segments; we do not derive the exact constrained path (a bang-singular problem) and instead use: cruise at u_{max} , enter and exit along the unconstrained arcs of effective size Δ_{sat} . Its cost is $\alpha\Delta^2/(2u_{\text{max}})(1+o(1))$ with the corner cost bounded by $2C^*(\Delta_{\text{sat}})$ — an explicitly declared approximation, conservative in the loss.

Proposition 9.5 (Ascent is always at the actuator bound). *Suppose the belief raises the argmin by Δ (the current level sits Δ too low), and call the deficit material when its integrated excess risk at any sub-saturated ascent exceeds the corner cost $2C^*(\Delta_{\text{sat}})$. During the ascent the system runs excess risk $\frac{c}{N} \sum_i [R_i(\mu_i(t)) - R_i(\mu_i^*)] \geq 0$, which decreases in the ascent rate for every t . Under the loss, the warp term charges the ascent nothing while the rate is held constant ($\ddot{\ell} = 0$); it charges only the entry and exit of the glide, a bounded one-time roughness cost that does not grow with Δ . Hence, for a material deficit, the optimal ascent runs at $\gamma = u_{\text{max}}$ until the level is restored, up to the bounded corner cost. For small deficits near the optimum — where the risk gradient is near zero — the soft mirror of the descent arc (theorem 9.4 with the roles of the endpoint costs exchanged) is the better move; saturating the clamp is only warranted when the risk at stake dominates the corners. The full ascent takes $\Delta/u_{\text{max}} - 2$ s per 10 ms of level — and this delay is irreducible: no admissible policy closes a level deficit faster. Consequently the level layer must be predictive where possible: a regime change detected early (a leading indicator of rising λ_i or σ_i) buys margin at the only speed the actuator allows.*

Proof. Risk excess decreases pointwise in the rate because a faster ascent reaches every intermediate level sooner and R_i is nonincreasing in the level. The corner cost is bounded exactly

as in Remark 9.3: entry and exit along unconstrained arcs cost at most $2C^*(\Delta_{\text{sat}})$, independent of Δ . The time bound is the actuator constraint $|\dot{\ell}| = |\bar{u} - \text{drift}| \leq u_{\text{max}} + O(10^{-4})$ from theorem 2.1. $\square$

Remark 9.4 (What asymmetry is and is not derived). Three asymmetries between raising and lowering the level follow from the loss; one popular claim does not, and we state both sides.

Derived. (1) *Stakes*: above the optimum the objective’s slope is $\alpha - \frac{c}{N} \sum_i g_i$, which is zero at the optimum and *bounded by* α , approaching α only well above it; below the optimum the slope reaches the occupation-density scale, which can exceed α by orders of magnitude. Sitting high is cheap; sitting low is not. (2) *Rates*: an ascent with a material deficit always saturates the clamp (theorem 9.5); a descent follows the unconstrained arc of theorem 9.4 whenever its peak rate $\frac{8\sqrt{3}}{27}\Delta^{3/4}(\alpha/r)^{1/4}$ is below the clamp — and this is slower for cheaper latency, $T^* \propto \alpha^{-1/4}$. The rate asymmetry is real but conditional: it comes from the latency-roughness tradeoff, not from any penalty on steady glides, and for large Δ both directions saturate. (3) *Decisions under uncertainty* (S7): delaying a descent by one second costs at most $\alpha\Delta$; delaying an ascent costs the integrated excess risk, which can be far larger. With a noisy argmin estimate the optimal trigger is therefore asymmetric: raise on weak evidence, lower only on statistically confident evidence.

Not derived. The loss contains no penalty on a steady glide and no rate limit other than the clamp; any policy that descends slower than theorem 9.4 prescribes is conservatism imported from outside this loss, not a consequence of it.

9.6 What remains of the general problem

Three couplings survive the decomposition, each with a bounded loss: in-event control (S3’s defect — grows with warp authority); estimation–control coupling (S7’s defect — shrinks with data); and the distributed-information gap: the theorem’s fast layer needs only the broadcast level, which tolerates the message delay because the level is slow (S6), but a formally optimal distributed policy for the transient case (one receiver disturbed, the fleet informed late) is not derived here. Team-decision problems of this kind have no general closed form (imported: the counterexample literature on distributed linear-quadratic control); the timescale argument bounds what the gap can cost, and section 12 records it as open.

10 The Resulting Controller

Theorem 9.1 licenses a controller with four layers. Nothing in the fast layer is new relative to the deployed system; the level layer and its broadcast are the new mechanism.

10.1 Layer 1: estimators (the belief state)

Per receiver, maintained by the telemetry already in production:

- Baseline $\hat{B}_i$: controller-matched exponential mean of the delay level. Its time constant matches the fast loop so that “deviation” means exactly the part of the delay the loop does not absorb.
- Deviation scale $\hat{\sigma}_i$ and the deviation histogram: estimates the bulk exceedance term of eq. (29); frozen during detected episodes so the bulk estimate is not contaminated by the tail it gates.
- Episode meter: rate $\hat{\lambda}_i$, height histogram $\hat{F}_i$, gap histogram $\hat{G}_i$, per-episode recovery $\hat{\rho}$. Every episode’s height is measured whether or not it underruns, so the risk curve is identified

at all levels without deliberately lowering the margin (S7).

- Drain validator: windowed margin minimum — the model-free check of the composed $\hat{R}_i$ (monitor of S2, S5).

10.2 Layer 2: the level program

At a slow cadence (seconds), the sender-side computes:

1. Compose $\hat{R}_i(\mu)$ from eq. (29) using layer-1 estimates; overlay the raw deviation-histogram exceedance where it has support (S2 monitor).
2. Solve eq. (33) for ℓ^* by bracketing (S8): find the level where the fleet-mean occupation density crosses α/c .
3. Rate-limit the output: descend along the quartic arc of theorem 9.4 (duration $2\sqrt{3}(r\Delta/\alpha)^{1/4}$, peak rate capped at $u_{\max}$ per Remark 9.3); ascend at $u_{\max}$ (theorem 9.5), triggered eagerly on regime-change detection (S1 monitor); gate descents on estimation confidence (S7, Remark 9.4).

The single number $\ell^*(t)$, plus per-receiver baseline offsets where backends differ, is broadcast to receivers. This is the only new protocol surface the theory requires.

10.3 Layer 3: fast loops (unchanged)

Each receiver tracks the broadcast level with the law of theorem 7.3 plus integral action eq. (25):

$$v = \kappa^2 e + K_0 \xi - \sqrt{2} \kappa u, \quad \kappa = (q/r)^{1/4}, \quad \zeta = 1/\sqrt{2},$$

with anti-windup at the warp clamp. This is the deployed second-order controller; Remark 7.2 maps the constants. By theorem 9.1(i) it is the optimum for its layer — no redesign is licensed by the joint problem.

10.4 Layer 4: event hygiene

During detected episodes: freeze the bulk estimators (layer 1), do not attempt heroic warp responses (S3: the authority is not there), record the episode's height, gap, and recovery for the risk curve. After the episode: let the redelivery burst restore the margin where the mechanism provides it; where recovery is warp-limited, the elevated compound risk during ρ is already priced by eq. (29).

10.5 Self-monitoring

Each simplification carries a live validity ratio (table 2). The controller can therefore report not only its decisions but the current size of the modeling error behind them. Two monitors deserve emphasis: the S3 ratio $u_{\max} \mu g(\mu)/R(\mu)$, because the entire decomposition stands on it, and the drain validator, because it checks the composed risk curve against reality without using the model.

11 Register of Simplifications

12 Limits of the Result

Stated as openly as the simplifications.

Table 2: All simplifications, their defects, and their monitors. Reference numbers use $u_{\max} = 5 \times 10^{-3}$, margins 5–100 ms, σ_i up to 1 ms wired, heights up to 1 s.

Simplification	Defect (definition and size)	Monitor
S1 stationary ergodic regime	loss during regime adaptation: switch rate $\times$ adaptation time $\times$ excess loss	short- vs long-window estimator disagreement
S2 additive risk evaluation	threshold smearing $O(\sigma_i/\text{height spread})$; duty cycle $O(\lambda \mathbb{E}H)$; zero-drain ramp (burst drain time)	composed $\hat{R}$ vs raw deviation histogram
S3 fast layer cannot move the indicator	absolute $\leq \alpha u_{\max} \mu$ ($\leq u_{\max}$ of the latency term); relative to the risk term: $u_{\max} \mu / x_0$, up to $\sim 5\%$ for the bulk part	ratio $u_{\max} \mu g(\mu) / R(\mu)$; anchor $\alpha u_{\max} \mu$
S4 frozen margin during episode	subsumed in S3	same
S5 one-step compounding	chain terms $O((\lambda^{\text{loc}} \rho)^2)$; gap–height covariance	episodes within ρ of predecessor; drain validator
S6 two-timescale separation	glide-concurrent episodes: margin error up to $\gamma H \leq 5$ ms	episodes stamped with glide rate
S7 certainty equivalence	$\frac{1}{2} J''(\mu^*) \text{Var}[\hat{\mu}^*]$; shrinks as $1/n$ with event count, inflated by clustering	confidence band on $\hat{\mu}^*$
S8 smooth, convex risk curve	subgradient bracket at atoms; compounding can break convexity in the clustered regime — program falls back to a 1-D scan, losing only the first-order characterization	$\hat{g}$ near operating point; second difference of scanned $\hat{J}$
S9 component independence (2nd-order accounting)	dropped cross-covariances; affects evaluations, not gains, not $\hat{R}$	baseline-drift vs event cross-correlation
S10 full-information law on the raw measurement	backend-dependent: measurement-noise passthrough $\int \kappa^2 / \mathcal{D} ^2 S_n$ (latency meas.); forfeited nonlinear gain vs pause leakage (margin meas.)	innovation histogram and spectrum; S_n per host
S11 zero tilts	forfeited loss $\sim \alpha^2 / 4q$; tilt scale $\alpha / 2q$ (microseconds)	spread of $c g_i(\mu_i)$ vs α
S12 white-noise surrogate in the fast-loop design	evaluation error $\int H ^2 (S_{\text{true}} - \sigma_w^2) d\omega / 2\pi$; the DC gain defect is repaired by integral action	innovation-spectrum flatness within κ
S13 common-mode loop response priced separately	omitted mean-warp roughness: the per-receiver roughness integral fed the common spectrum; drift-rate part $\leq 10^{-16} r$	cross-receiver coherence of loop innovations; fleet-average warp-rate spectrum vs commanded $\ddot{\ell}$

(L1) **Warp authority.** The decomposition stands on S3. If $u_{\max}$ were raised by an order of magnitude or more, in-event control would become worthwhile, the indicator would couple to the fast layer, and the correct formulation is a free-boundary dynamic program that this document does not solve. The S3 monitor quantifies the approach to that regime.

(L2) **Risk aversion.** The loss charges mean silence time. A listener may care more about the

worst minute than the average second. A variance-sensitive or quantile objective re-imports the temporal law (clustering) into the *objective* — see Remark 5.3 — and the level program stops being a one-dimensional convex program in a single scalar.

- (L3) **Session supervision.** Aborts and restarts (Remark 4.2) are outside the loss. In particular, implementations in which playout stalls instead of inserting silence violate Convention 2.2: a stall converts an episode into a schedule slip whose recovery is warp-limited (Assumption 3.5(b)) and whose transient can trip supervision thresholds. The convention–implementation gap is a real difference, not a modeling detail, and must be closed on the implementation side or modeled explicitly.
- (L4) **Distributed transients.** The optimal reaction of the undisturbed majority to one disturbed receiver, under message delay, is not derived (section 9.6). The forfeited value is bounded by the disturbance rate times the reconvergence transient.
- (L5) **Lost content.** Episodes whose content is lost rather than delivered late cause silence that no margin prevents; that channel belongs to forward error correction, not to latency control, and is additive to the loss studied here.
- (L6) **Adversarial or trending regimes.** S1 covers slow regime switching with re-estimation. Systematically trending networks (a slow decay toward failure) are tracked by the belief but never anticipated beyond the leading indicators of theorem 9.5.

A Imported Results

A.1 Campbell–Mecke formula

For a stationary marked point process on $\mathbb{R}$ with intensity λ and Palm mark distribution F , and any nonnegative measurable function f of the mark,

$$\mathbb{E}\left[\sum_{k: T_k \in [0, T]} f(H_k)\right] = \lambda T \int f \, dF.$$

Reference: Daley and Vere-Jones, *An Introduction to the Theory of Point Processes*, Vol. II, Ch. 13 (Palm theory); the ergodic form used in theorem 5.5 is the corresponding ratio limit theorem. Note what is *not* required: no Poisson assumption, no independence between points — stationarity suffices, which is why clustering does not move the mean in eq. (16).

A.2 Rice’s formula

For a stationary, almost surely continuously differentiable Gaussian process X with $\text{Var}[X] = \sigma_0^2$ and $\text{Var}[\dot{X}] = \sigma_1^2$, the mean rate of upcrossings of level z is

$$N^+(z) = \frac{\sigma_1}{2\pi\sigma_0} \exp\left(-\frac{z^2}{2\sigma_0^2}\right).$$

Reference: Rice (1944); Leadbetter, Lindgren, Rootzén, *Extremes and Related Properties of Random Sequences and Processes*, Ch. 7. The same reference (Ch. 7, sojourn results) covers the ergodic occupation–crossing ratio used alongside it: for a stationary process with finitely many level crossings per unit time, the mean duration of a visit above z equals the stationary exceedance probability divided by the upcrossing rate. The sojourn computation of theorem 5.4 combines exactly these two imports.

A.3 Gaussian tail independence

Lemma A.1. *Let (Z_1, Z_2) be bivariate standard normal with correlation $\varsigma < 1$. Then $\lim_{z \rightarrow \infty} \mathbb{P}(Z_2 > z \mid Z_1 > z) = 0$.*

Proof. For $\varsigma \leq 0$, Slepian's comparison inequality (imported; Slepian 1962) gives $\mathbb{P}(Z_1 > z, Z_2 > z) \leq \mathbb{P}(Z_1 > z)\mathbb{P}(Z_2 > z)$, so the conditional probability is at most the unconditional one, $\bar{\Phi}(z) \rightarrow 0$; assume therefore $0 < \varsigma < 1$. Condition on $Z_1 = x$: $Z_2 \sim \mathcal{N}(\varsigma x, 1 - \varsigma^2)$, so $\mathbb{P}(Z_2 > z \mid Z_1 = x) = \bar{\Phi}((z - \varsigma x)/\sqrt{1 - \varsigma^2})$. Split the conditioning event at $x = z(1 + \epsilon)$ for a fixed $\epsilon > 0$. For $x \leq z(1 + \epsilon)$: $(z - \varsigma x)/\sqrt{1 - \varsigma^2} \geq z(1 - \varsigma(1 + \epsilon))/\sqrt{1 - \varsigma^2} \rightarrow \infty$ when $\epsilon < (1 - \varsigma)/\varsigma$, so the conditional probability tends to 0 uniformly on this range. For the range $x > z(1 + \epsilon)$, its share of the conditioning mass is $\bar{\Phi}(z(1 + \epsilon))/\bar{\Phi}(z) \rightarrow 0$ by the Mills ratio $\bar{\Phi}(t) = \varphi(t)/t(1 + O(t^{-2}))$, since $\varphi(z(1 + \epsilon))/\varphi(z) = e^{-z^2(2\epsilon + \epsilon^2)/2} \rightarrow 0$. Both contributions vanish. $\square$

Reference: Sibuya (1960). This is the mechanism behind theorem 5.4: a Gaussian-copula process cannot hold an extreme level; a pause can and does.

A.4 Other imports

The verification theorem for average-cost linear-quadratic control (theorem 7.3): Fleming and Soner, *Controlled Markov Processes and Viscosity Solutions*, or any standard treatment of ergodic LQG. The spectral representation of stationary processes (theorem 7.2): Wiener–Khinchin/Bochner, standard. The distributed-control counterexamples referenced in section 9.6: Witsenhausen (1968).

B Notation

Symbol	Meaning	Units	Defined
s	stream position	s (audio)	Definition 2.1
$\tau(s)$	emission time of s	s	Definition 2.1
$a_i(s)$	arrival time at receiver i	s	Definition 2.2
$D_i(s)$	delivery delay	s	Definition 2.2
$p_i(s), s_i(t)$	playout schedule and its inverse	s	Definition 2.3
$u_i, u_{\max}$	warp; warp clamp (5×10^{-3})	—	Definition 2.3
$v_i = \dot{u}_i$	warp rate (control input)	s^{-1}	eq. (19)
$L_i, \bar{L}, \tilde{L}_i$	latency; fleet mean; deviation	s	Definitions 2.4 and 6.1
w_i	clock drift disturbance	—	theorem 2.1
m_i	margin	s	Definition 2.6
B_i, ε_i, P_i	baseline, diffusion, pauses	s	Assumption 3.1
σ_i	diffusion scale	s	Assumption 3.3
$(T_{i,k}, H_{i,k})$	episode start, height	s	Assumption 3.4
λ_i, F_i, G_i	episode rate; height law; gap law	$s^{-1}, —, —$	Assumption 3.4
$\rho_{i,k}, \chi_{i,k}$	recovery time; consumed margin	s	Definition 3.1 and eq. (30)
x_i	deviation $\varepsilon_i + P_i$	s	eq. (26)
α, q, c, r	loss weights	see Remark 4.1	Definition 4.2
$\kappa = (q/r)^{1/4}$	fast-loop bandwidth	s^{-1}	theorem 7.3
ℓ, ℓ^*	latency level; its optimum	s	section 9.3
μ_i	margin level (backend-conditional)	s	eq. (27)
$R_i(\mu), g_i(\mu)$	risk curve; occupation density	—, s^{-1}	eq. (28) and Definition 8.1
γ	level glide rate	—	section 9.5
$\mathcal{D}(s)$ (Laplace)	$s^2 + \sqrt{2}\kappa s + \kappa^2$	—	theorem 7.3
σ_w^2	white-noise intensity (wander + measurement)	s^2/s	section 7.3
$\Lambda(t, \omega)$	pointwise loss rate	u/s	eq. (13)
e, u, v	fast-loop error, warp, input (one coordinate)	s, —, s^{-1}	eq. (19)
ξ, K_0, T_I	integral state, gain, settling time	s^2, s^{-3}, s	eq. (25)
$\delta(g)$	residual margin depression at gap g	s	eq. (30)
Π	centering matrix $I - \mathbf{1}\mathbf{1}^\top/N$	—	theorem 7.5
$\bar{\beta}$	baseline slope bound	—	Assumption 3.2
x_0, C	diffusion tail scale and constant	s, —	Assumption 3.3
$\Delta, \Delta_{\text{sat}}$	level change; saturation threshold	s	theorem 9.4
$T^*, \gamma_{\text{pk}}, C^*$	descent duration, peak rate, cost	s, —, u	theorem 9.4
$J, J_{\text{diff}}, J_{\text{mean}}$	loss and its decomposition parts	u/s	Definition 4.2 and theorem 9.1
$\tau_{\hat{B}}$	baseline filter time constant	s	section 8.1
n_i, N_i	measurement error (time domain, transform)	s	Definition 2.7 and Convention 7.1
$\tilde{D}$	transform of the deviation x_i	s	Convention 7.1
$\hat{\theta}$	belief state (estimator outputs)	—	Definition 8.2
e_i^{loop}	fast-loop tracking error	s	eq. (26)
b_i	baseline offsets in the level objective	s	theorem 8.2
λ_i^{loc}	local (in-cluster) episode rate	s^{-1}	Simplification 5
$\bar{\Phi}, \varphi$	standard normal tail, density	—	standard
ϱ	average-cost constant	u/s	theorem 7.3
ς	correlation (tail-independence lemma)	—	theorem A.1
$\Delta_S(\pi)$	evaluation error of the decomposition	u/s	theorem 9.1